Self-Normalized Martingale AIPW/DML for Adaptive Experiments with Logged Propensities

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Abstract

We study prespecified-horizon Wald inference for the superpopulation average treatment effect (ATE) in sequentially adaptive randomized experiments with logged assignment probabilities. Because the treatment propensity π_t may keep changing, the predictable quadratic variation of standard AIPW/DML scores can remain random across replications, so normal approximations that rely on a deterministic long-run variance can be ill-calibrated conditionally on the realized variance regime. (Here "DML-style" refers to cross-fitting the outcome regressions only; propensities are treated as known/logged.) Under standard causal assumptions (SUTVA and sequential randomization), i.i.d. arrivals, executed overlap on a prespecified scored set (e.g., units after an initial burn-in), and finite fourth-moment and nuisance-stability conditions (with a $(2+\delta)$-moment alternative; see Remark 3.12), we emphasize two additional *auditable* pipeline requirements: (i) the platform logs, for every unit, the *executed* assignment probability used to randomize A_t; and (ii) the nuisance regressions used to score unit t are constructed predictably using only past data. These conditions make the centered AIPW score increments an exact martingale difference sequence. We then apply self-normalized martingale limit theory to show that the usual Studentized statistic—with variance estimated by realized quadratic variation—is asymptotically $\mathcal{N}(0,1)$ at the prespecified horizon even when no deterministic variance limit exists. Valid coverage does not require nuisance consistency; under L^2 convergence we obtain oracle equivalence, and a conditional second-moment decomposition provides an oracle benchmark for precision. We conclude with a concrete logging-and-fitting protocol that separates auditable pipeline conditions from substantive assumptions; the resulting intervals are *not* anytime-valid and should not be used under optional stopping.

1 Introduction

Adaptive randomized experiments—including response-adaptive clinical trials, contextual bandits, and large-scale platform experimentation systems—update assignment probabilities as data accrue to balance learning and deployment (Kasy and Sautmann, 2021; Zhang et al., 2020). In practice, however, many platforms still require conventional end-of-study reporting for classical causal estimands such as the superpopulation average treatment effect (ATE) computed once at a prespecified horizon. Hereafter, we denote by π_t the executed assignment probability used to randomize unit t after applying any platform guardrails, as recorded in the experiment log. In this paper, we study fixed-horizon Wald inference for the standard logged-propensity AIPW/DML estimator, the sample average of doubly robust pseudo-outcomes scored using these logged propensities. Under adaptive assignment, the propensity process $\{\pi_t\}$ is itself data-dependent, so the predictable quadratic variation of the AIPW/DML score increments can remain replication-random and need not converge to a single deterministic long-run variance target.

Some end-of-study Wald arguments for AIPW/A2IPW (and related DML-style estimators) under adaptivity proceed via Slutsky steps that rely on a deterministic variance target for the predictable quadratic variation, often enforced through stabilization-type or design-stability conditions on assignment probabilities and/or average conditional variances (Hadad et al., 2021; Zhan et al., 2021; Kato et al., 2020; Cook et al., 2024; Li and Owen, 2024; Sengupta et al., 2025). On modern platforms, however, the policy can keep reacting to noisy intermediate estimates. Clipping and guardrails can activate intermittently and batch updates can induce regime switches (Kasy and Sautmann, 2021; Zhang et al., 2020). When this happens, a Wald statistic normalized using a single deterministic variance target can be systematically miscalibrated conditional on the realized variance regime, even if marginal coverage appears close to nominal.

Our route is different. We treat the adaptive assignment policy as given but assume that the platform logs the executed propensity used to randomize each unit and that nuisance regressions used for AIPW/DML scoring are fit predictably using only past data. Related work emphasizes that careful use of the logging policy and past-only fitting is central for post-adaptive inference (Bibaut et al., 2021; Kato et al., 2021; Cook et al., 2024). Under these auditable conditions, the centered AIPW score increments form an exact martingale difference sequence, and we obtain fixed-horizon Wald inference by studentizing with realized quadratic variation along the realized propensity path. This yields asymptotic $\mathcal{N}(0,1)$ calibration without requiring the predictable quadratic variation to converge to a deterministic long-run variance target.[1]

Contributions.

- **Auditable martingale scoring.** We formalize a logging/predictability contract, logged executed propensities and predictable nuisance fitting, under which centered AIPW/DML score increments form an exact martingale difference sequence (Lemma 5.3).
- **Fixed-horizon self-normalized Wald inference.** We prove that the usual Studentized statistic, with variance estimated by realized quadratic variation, is asymptotically $\mathcal{N}(0,1)$ at a prespecified horizon even when no deterministic long-run variance limit exists (Theorem 5.14).
- **Feasible studentization.** We show that the standard plug-in studentizer used in practice consistently

[1] We do not claim anytime-valid or optional-stopping guarantees.

estimates realized quadratic variation, so the feasible Wald interval inherits the same fixed-horizon validity (Proposition 4.11).

- **Oracle benchmarking and nuisance-learning effects.** We provide a conditional second-moment decomposition yielding an oracle precision benchmark and isolate a nonnegative augmentation term capturing variance inflation from nuisance error (Proposition 5.8). Under weighted L^2 convergence, the feasible statistic is asymptotically oracle-equivalent (Theorem C.4).

- **Operational protocol and simulations.** We give an audit-ready logging-and-fitting protocol (Algorithm 1 and Appendix F) and simulations illustrating conditional miscalibration when normalizing by deterministic variance targets and the corresponding robustness of quadratic-variation studentization (Section 6).

Our analysis assumes that the experiment log records, for each unit, the executed assignment probability actually passed to the randomization device (Assumption 3.2) and that all nuisance estimates used at time t are measurable with respect to the past σ-field $\mathcal{F}_{t-1}$ (Assumption 4.5). The resulting Wald intervals are justified at a prespecified horizon and are not intended for optional stopping or continuous monitoring. Readers who need time-uniform validity should use confidence sequences or time-uniform CLT analogues (Howard et al., 2021; Waudby-Smith et al., 2024).

To avoid conditioning on a hypothetical deterministic variance target, we studentize by the realized quadratic variation along the realized assignment path.[2] Under predictable nuisance fitting and logged executed propensities, the AIPW/DML score becomes a martingale difference sequence (MDS). We impose an auditable data contract that is natural for modern experimentation platforms: (i) the platform logs, unit by unit, the executed assignment probability π_t passed to the randomization device; (ii) the nuisance fits used to score unit t are predictable given $\mathcal{F}_{t-1}$ (Assumption 4.5); and (iii) executed overlap holds on the prespecified scored set, so π_t is uniformly bounded away from 0 and 1. Under these checkable conditions, the centered AIPW/DML score increments form an exact martingale difference sequence (Lemma 5.3), yielding finite-sample conditional unbiasedness and making the experiment's uncertainty regime analyzable through its quadratic variation.

The subsequent sections are structured as follows. Section 2 reviews related work. Section 3 states the model and assumptions. Section 4 presents the estimator and auditable implementation details. Section 5 develops the main theoretical results and Section 6 reports simulations. The appendices collect supporting limit-theory background, additional results and proofs, and an operational logging protocol.

2 Related Work

Inference after adaptive data collection

A growing literature studies inference with adaptively collected data, where observations are generated under an evolving information set and classical i.i.d. arguments do not directly apply. Early econometric work by Hahn et al. (2011) highlighted how propensity information can be leveraged for inference in sequential

[2] Our theorems establish marginal fixed-horizon coverage. Conditional calibration across realized variance regimes is illustrated in Design A and is not asserted generically.

designs. More recent general frameworks derive asymptotic representations for sequential decisions and adaptive experiments under broad conditions (Hirano and Porter, 2023). In the contextual-bandit and adaptive-experiment literature, fixed-horizon inference has also been developed via weighted M-estimation and related estimating-equation arguments (Zhang et al., 2021). Our focus is narrower but operationally central for experimentation platforms: fixed-horizon ATE reporting with logged executed propensities and predictable AIPW/DML scoring. In particular, we target settings where the predictable quadratic variation remains random across replications, so that deterministic-variance normalizations can be conditionally miscalibrated. General background on response-adaptive randomization and bandit-style designs can be found in Rosenberger and Lachin (2016) and Villar et al. (2015).

Stabilization, adaptive weighting and batching

In adaptive experimentation and off-policy evaluation, evolving propensities can create heavy tails and regime-dependent uncertainty, motivating variance-control strategies. In policy evaluation, Hadad et al. (2021) and Zhan et al. (2021) develop adaptive weighting schemes for augmented IPW/DR scores to obtain asymptotically normal t-statistics. For post-contextual-bandit inference, Bibaut et al. (2021) proposes stabilized doubly robust constructions that estimate conditional scale components using only past data. Another route restricts the data-collection design—for example through batching—to recover classical CLTs for bandit estimators (Zhang et al., 2020). In contrast, we keep the standard logged-propensity AIPW/DML estimator unchanged and do not require batching or adaptive reweighting. Instead, we rely on a martingale score representation and studentization by realized quadratic variation.

Design stability and deterministic variance-limit CLTs

A further line of work derives fixed-horizon CLTs for IPW/AIPW-type ATE estimators under explicit design stability conditions ensuring that inverse-propensity averages and/or average conditional variances converge to nonrandom limits, yielding conventional deterministic asymptotic variances for Wald reporting (Sengupta et al., 2025). Related perspectives arise when one engineers assignment rules (or batchwise designs) to target efficiency or precision for A2IPW/DML-style estimators (Kato et al., 2020; Li and Owen, 2024; Cook et al., 2024). More generally, the adaptive-experiment literature emphasizes subtleties around what "the logging policy" means operationally when platforms implement clipping, guardrails, or algorithmic randomness (Kato et al., 2021). Our contribution is complementary: we assume the executed propensity actually used to randomize each unit is logged and we avoid requiring convergence of the predictable quadratic variation to a deterministic limit by using realized quadratic variation as the normalizer.

Self-normalized martingale theory for quadratic-variation studentization

Our fixed-horizon Wald statistic is a self-normalized martingale functional. Classic references for martingale CLTs and self-normalized processes include Hall and Heyde (1980) and de la Peña et al. (2009), as well as the survey Shao and Wang (2013). Modern probability theory provides refined asymptotic and nonasymptotic control for self-normalized martingales, including Berry–Esseen bounds (Fan and Shao, 2018), Cramér-type moderate deviations (Fan et al., 2019), and concentration inequalities (Bercu and Touati, 2019). We apply

this theory to AIPW/DML score increments under adaptive assignment, using realized quadratic variation to obtain a fixed-horizon $\mathcal{N}(0,1)$ approximation without a deterministic variance limit.

Anytime-valid and time-uniform alternatives

The results in this paper are prespecified-horizon and do not provide optional-stopping guarantees. When inference must remain valid under continuous monitoring or data-dependent stopping, time-uniform methods based on test supermartingales and confidence sequences are appropriate (Howard et al., 2021; Waudby-Smith et al., 2024). Related time-uniform tools also appear alongside fixed-time inference in adaptive-experiment work (Cook et al., 2024; Kato et al., 2020; Waudby-Smith and Ramdas, 2024). We focus instead on conventional fixed-horizon reporting with realized quadratic-variation studentization (see Remark 5.16 for additional discussion).

3 Model and Assumptions

3.1 Propensities and Logged Assignment Rule

Consider a stream of experimental units indexed by $t = 1, \ldots, n$. For each unit, covariates X_t are observed before assignment. The platform computes an assignment probability $\pi_t \in (0,1)$. Then, randomizes $A_t \in \{0,1\}$ using π_t. Finally, observes an outcome Y_t.

We observe and store $Z_t := (X_t, A_t, Y_t, \pi_t)$ in time order. Let

$$\mathcal{F}_t := \sigma(Z_1, \ldots, Z_t), \qquad \mathcal{F}_0 = \{\emptyset, \Omega\}, \tag{3.1}$$

be the data filtration. Let

$$\mathcal{G}_t := \sigma(\mathcal{F}_{t-1}, X_t, \pi_t) \tag{3.2}$$

be the σ-field immediately before randomization at time t, after the platform has computed the executed assignment probability π_t. The policy may be arbitrarily adaptive or randomized. The fixed-horizon validity results below require only that the realized executed propensity used to randomize A_t is recorded in the log (Assumption 3.2).

Definition 3.1 (Adaptive logged executed propensity)**.** At time t, after observing $(\mathcal{F}_{t-1}, X_t)$ and any algorithmic randomness used to form the propensity, the platform records the executed assignment probability $\pi_t \in (0,1)$ and draws treatment according to

$$A_t \mid \mathcal{G}_t \sim \text{Bernoulli}(\pi_t),$$

where $\mathcal{G}_t$ is the pre-treatment σ-field in (3.2). If π_t is deterministic given $(\mathcal{F}_{t-1}, X_t)$, then $\pi_t = \mathbb{P}(A_t = 1 \mid \mathcal{F}_{t-1}, X_t)$. If π_t is produced using additional exogenous randomness, then $\mathbb{P}(A_t = 1 \mid \mathcal{F}_{t-1}, X_t) = \mathbb{E}[\pi_t \mid \mathcal{F}_{t-1}, X_t]$ while still $\mathbb{P}(A_t = 1 \mid \mathcal{G}_t) = \pi_t$. We treat any such platform-side randomness as exogenous, i.e., independent of $(Y_t(0), Y_t(1))$ conditional on $(\mathcal{F}_{t-1}, X_t)$.

Assumption 3.2 (Logging integrity)**.** For each t, the logged propensity π_t is $\mathcal{G}_t$-measurable, takes values in

$(0, 1)$, and equals the probability passed to the randomization device that generated A_t, i.e.,

$$\mathbb{P}(A_t = 1 \mid \mathcal{G}_t) = \pi_t \quad \text{a.s.}$$

Equivalently, $A_t \mid \mathcal{G}_t \sim \text{Bernoulli}(\pi_t)$ with the logged π_t.

3.2 Audit diagnostics and data contract

Remark 3.3 (Auditing logged propensities). Assumption 3.2 requires that, conditional on the platform history $\mathcal{G}_t$ (including the logged probability π_t), the treatment assignment satisfies $A_t \mid \mathcal{G}_t \sim \text{Bernoulli}(\pi_t)$. This is a design-stage property. It holds only if the log records the probability that was actually passed to the randomization device. A basic diagnostic is calibration of A_t against π_t. For any bin $B \subset (0, 1)$ with many observations, let

$$N_B := \sum_{t=1}^{n} \mathbb{1}\{\pi_t \in B\}, \qquad \overline{A}_B := \frac{1}{N_B} \sum_{t:\pi_t \in B} A_t, \qquad \overline{\pi}_B := \frac{1}{N_B} \sum_{t:\pi_t \in B} \pi_t.$$

In practice, restrict attention to bins with $N_B \geq N_{\min}$ for a user-chosen $N_{\min} \geq 1$ (or merge empty/small bins). Under correct logging, $\overline{A}_B$ should be close to $\overline{\pi}_B$ up to martingale sampling variability; under adaptivity, dependence can matter, so formal testing can be based on martingale concentration/self-normalized methods for the MDS $(A_t - \pi_t)$. These checks can detect severe mis-logging or implementation bugs, but they cannot certify correct logging or validate the causal assumptions in Assumptions 3.5–3.7.

Remark 3.4 (Timing and measurability). The experiment unfolds in the order (1) observe X_t; (2) the platform computes and logs an executed propensity π_t as a function of $(\mathcal{F}_{t-1}, X_t)$; (3) treatment is drawn according to $A_t \mid \mathcal{G}_t \sim \text{Bernoulli}(\pi_t)$; and (4) the outcome Y_t is revealed and logged. In particular, π_t is $\mathcal{G}_t$-measurable, while (A_t, Y_t) are $\mathcal{F}_t$-measurable. This timing underlies the predictability requirement in Section 4 and the data-contract protocol in Appendix F.

Step	Object	Measurability / check
1	Observe covariates X_t	observed at t (pre-treatment)
2	Choose/log π_t	π_t must be $\mathcal{G}_t$-measurable; log the executed random-number generator (RNG) probability
3	Randomize A_t	$A_t \mid \mathcal{G}_t \sim \text{Bernoulli}(\pi_t)$ (Assumption 3.2)
4	Observe outcome Y_t	realized after A_t
5	Fit/update nuisances for $t+1$	nuisance used at $t+1$ must be $\mathcal{F}_t$-measurable

Table 1: Timing and measurability. The key analytical requirement is that the nuisance used to score unit t is $\mathcal{F}_{t-1}$-measurable (Assumption 4.5).

3.3 Potential Outcomes and Target Parameter

We adopt the potential outcomes framework with the usual no interference and consistency conventions.

Assumption 3.5 (Consistency and no interference; SUTVA)**.** For each unit t, there are well-defined potential outcomes $(Y_t(0), Y_t(1))$ that depend only on the assignment of unit t itself (see Rubin, 1980). The observed

Symbol	Meaning
n, t	fixed horizon; unit index $t = 1, \ldots, n$
Z_t	logged record (X_t, A_t, Y_t, π_t)
$\mathcal{F}_t, \mathcal{G}_t$	post-outcome history; pre-treatment info (3.1)–(3.2)
π_t	logged executed propensity; $\mathbb{P}(A_t = 1 \mid \mathcal{G}_t) = \pi_t$ (Ass. 3.2)
$Y_t(a), Y_t$	potential outcomes; observed $Y_t = A_t Y_t(1) + (1 - A_t) Y_t(0)$
$m_a^\star$	true regression $x \mapsto \mathbb{E}[Y(a) \mid X = x]$
$\widehat{m}_{t-1,a}$	predictable nuisance used at time t (Ass. 4.5)
$\phi_t(m, \pi)$	AIPW/DR pseudo-outcome evaluated at propensity π (4.1)
$\widehat{\phi}_t, \widehat{\theta}$	scored pseudo-outcome $\widehat{\phi}_t = \phi_t(\widehat{m}_{t-1}, \pi_t)$ and estimator (4.3)
$\theta_0, \widehat{\theta}$	ATE (3.3); estimator (4.3)
$\xi_t, S_\mathcal{T}$	increment $\xi_t = \widehat{\phi}_t - \theta_0$; sum $S_\mathcal{T} = \sum_{t \in \mathcal{T}} \xi_t$
$\mathcal{T}, n_{\text{eff}}$	scored index set; $n_{\text{eff}} := \lvert\mathcal{T}\rvert$
$Q_\mathcal{T}, V_\mathcal{T}^2, \widehat{V}$	realized/predictable QV (5.3); studentizer (4.4)

Table 2: Summary of notation.

outcome satisfies

$$Y_t = Y_t(A_t) = A_t Y_t(1) + (1 - A_t) Y_t(0) \qquad \text{a.s.}$$

The target parameter is the average treatment effect (ATE) in the superpopulation:

$$\theta_0 := \mathbb{E}\big[Y(1) - Y(0)\big], \tag{3.3}$$

where $(X, Y(0), Y(1))$ denotes a generic draw from the superpopulation described in Assumption 3.6. Assumption 3.5 is standard. It rules out spillovers or network effects, which require separate methods.

3.4 Assumptions

We maintain the following assumptions throughout.

Assumption 3.6 (Superpopulation arrivals; no selection)**.** The sequence $\{(X_t, Y_t(0), Y_t(1))\}_{t=1}^n$ is independent across t and identically distributed with generic draw $(X, Y(0), Y(1))$. In particular, the adaptive policy affects only treatment assignment, not which units/covariates arrive.

We write all expectations with respect to this superpopulation draw, so $\theta_0 = \mathbb{E}[Y(1) - Y(0)]$ and $\tau(x) := \mathbb{E}[Y(1) - Y(0) \mid X = x]$ satisfy $\theta_0 = \mathbb{E}[\tau(X)]$. As noted in Remark 5.7, the proofs only require the conditional mean-stationarity condition $\mathbb{E}[\tau(X_t) \mid \mathcal{F}_{t-1}] = \theta_0$.

Assumption 3.7 (Sequential randomization / no anticipation)**.** $A_t \perp\!\!\!\perp (Y_t(0), Y_t(1)) \mid \mathcal{G}_t$, where $\mathcal{G}_t$ is the pre-treatment σ-field in (3.2). Equivalently, conditional on the pre-treatment information (including the realized executed propensity π_t), the treatment draw is independent of the potential outcomes.

Assumption 3.8 (Executed overlap on scored units)**.** There exists $\varepsilon > 0$ such that for every time index whose score is included in the final estimator (i.e., $t \in \mathcal{T}$ in Section 4),

$$\varepsilon \le \pi_t \le 1 - \varepsilon \qquad \text{a.s.} \tag{3.4}$$

A sufficient operational mechanism is to enforce clipping of the executed propensity before randomization on all scored units.

Assumption 3.9 (Moment conditions)**.** For each $a \in \{0, 1\}$, $\mathbb{E}[|Y(a)|^4] < \infty$.

Assumption 3.10 (Nuisance stability)**.** There exists a finite constant $C < \infty$ such that for each $a \in \{0, 1\}$,

$$\sup_{n \geq 1} \sup_{t \in \mathcal{T}} \mathbb{E}[|\widehat{m}_{t-1,a}(X_t)|^4] \leq C.$$

In unbounded-outcome settings, a sufficient alternative is to truncate/clamp $\widehat{m}_{t-1,a}(X_t)$ at a large threshold; see Remark 3.11.

Remark 3.11 (How to enforce Assumption 3.10). If outcomes are known to be bounded, say $Y_t \in [-B, B]$, we can enforce Assumption 3.10 by clipping $\widehat{m}_{t-1,a}(x)$ to $[-B, B]$ (or slightly wider) for each a. For unbounded outcomes (as in our Gaussian simulations), one may instead truncate at a large threshold or use robust regression; this is a technical device to control moments and does not change the target estimand.

Remark 3.12 (Weaker moment conditions are possible but are not needed for this note). Self-normalized martingale CLTs only require finite $(2 + \delta)$ moments of the score increments. We impose fourth moments to keep the quadratic-variation equivalence proof fully transparent.

Remark 3.13 (Design-stage, analysis-stage and auditable conditions). Our validity results combine (i) substantive causal/model assumptions and (ii) pipeline conditions that can be checked from logs and stored artifacts.

1. *Substantive assumptions.* SUTVA/no interference (Assumption 3.5) and sequential randomization with a stable target estimand (Assumption 3.7), together with i.i.d. arrivals/no selection (Assumption 3.6) or an explicit mean-stationarity alternative (Remark 5.7).

2. *Auditable design-time requirements.* Logged executed propensities (Assumption 3.2) and executed overlap on the scored units (Assumption 3.8), which must be enforced at assignment time if needed (e.g., by clipping π_t).

3. *Auditable analysis-time requirements.* A prespecified (deterministic) scored set $\mathcal{T}$ (Assumption 4.8) and predictable nuisance construction (Assumption 4.5), which rule out "peeking" when fitting the regressions used to score each unit.

4. *Regularity conditions.* Moment and stability bounds (Assumptions 3.9 and 3.10) and variance growth (Assumption 5.11) needed to apply martingale limit theory.

Finally, all confidence intervals are fixed-horizon: they should be reported only at the prespecified sample size and are generally invalid under optional stopping (Remark 5.16 and Appendix F).

4 Predictable AIPW/DML Estimation

4.1 The Doubly Robust Score

Let $m_a^\star(x) := \mathbb{E}[Y(a) \mid X = x]$ and write $m^\star := (m_0^\star, m_1^\star)$. For any candidate $m = (m_0, m_1)$ and any $\pi \in (0, 1)$, define the usual augmented inverse-propensity weighted (AIPW) / doubly robust (DR) score

(Robins et al., 1994; Bang and Robins, 2005) (pseudo-outcome)

$$\phi_t(m,\pi) := \frac{A_t}{\pi}\big(Y_t - m_1(X_t)\big) - \frac{1-A_t}{1-\pi}\big(Y_t - m_0(X_t)\big) + m_1(X_t) - m_0(X_t). \tag{4.1}$$

When evaluating at the logged executed propensity, we write $\phi_t(m) := \phi_t(m,\pi_t)$ for brevity. The quantity $\phi_t(m,\pi_t)$ is observable given (X_t, A_t, Y_t, π_t) and a supplied regression pair m.

For the oracle regression $m^\star$, one can decompose

$$\phi_t(m^\star) = \tau(X_t) + \frac{A_t}{\pi_t}\big(Y_t(1) - m_1^\star(X_t)\big) - \frac{1-A_t}{1-\pi_t}\big(Y_t(0) - m_0^\star(X_t)\big). \tag{4.2}$$

The centered increment $\phi_t(m^\star) - \theta_0$ has mean zero and is the object governed by the martingale CLT.

Remark 4.1 (Terminology). The estimating function $\phi_t(m,\pi_t)$ in (4.1) has the classical doubly robust/AIPW form. When evaluated at the true regression functions $m^\star$ (and the executed propensities), the centered score $\phi_t(m^\star,\pi_t) - \theta_0$ coincides with the efficient influence function for the ATE in the corresponding i.i.d. model with known propensity (see, e.g., (Hahn, 1998; Tsiatis, 2006)); for general m, it is simply an estimating function with the same doubly robust form. In off-policy evaluation and contextual-bandit settings, essentially the same form appears as a doubly robust score for value estimation (Dudik et al., 2011, 2014).

Remark 4.2 (Multi-arm extensions). The results extend directly to $K > 2$ arms by replacing the scalar propensity with a probability vector and using the corresponding multivariate AIPW score. For simplicity, we present the binary case.

4.2 Forward Cross-Fitting

Standard cross-fitting partitions data into folds and estimates nuisances on held-out folds. Under adaptivity, we must respect time: nuisance estimates used at time t must be constructed from data strictly before t.

Definition 4.3 (Forward cross-fitting)**.** Partition indices $\{1,\dots,n\}$ into K contiguous blocks $I_1,\dots,I_K$. The partition is deterministic and fixed prior to data collection. For each block $k \geq 2$:

1. Estimate nuisance functions $(\widehat{m}_0^{(-k)}, \widehat{m}_1^{(-k)})$ using only data from blocks $I_1,\dots,I_{k-1}$.
2. For each $t \in I_k$, evaluate the score using these estimates and the logged propensity π_t.

For $t \in I_1$, use a pilot estimate or exclude I_1 from inference.

Remark 4.4 (Common pitfall). A common pitfall is to use i.i.d.-style sample splitting or cross-fitting patterns that inadvertently allow information from unit t (or future units) to enter the nuisance regression used to score unit t. In adaptive experiments, such "leakage" violates Assumption 4.5 and can break the martingale difference property in Lemma 5.3. As a result, the Studentized Wald statistic built from leaky scores generally lacks the fixed-horizon guarantee of Theorem 5.14: it may still work in some finite-sample designs, but its validity is no longer ensured by the martingale argument. Appendix F gives an auditable implementation pattern (forward cross-fitting) that enforces predictability.

Assumption 4.5 (Predictable nuisance construction)**.** For each scored time $t \in \mathcal{T}$ and treatment arm $a \in \{0,1\}$, the regression function $\widehat{m}_{t-1,a}$ used in the score for unit t is $\mathcal{F}_{t-1}$-measurable. Equivalently, conditional on the past $\mathcal{F}_{t-1}$, the function $\widehat{m}_{t-1,a}$ does not depend on (A_t, Y_t) or on any future data.

Assumption 4.5 is the formal predictability requirement ('no-peeking", i.e., no use of contemporaneous or future outcomes) that makes the scored pseudo-outcomes a martingale difference sequence, and predictable fits are not based on non-predictable ("leaky") scores (see Lemma 4.6 below). The measurability here refers to the fitted function object $\widehat{m}_{t-1,a}$; it may then be evaluated at the current covariate X_t to form $\widehat{m}_{t-1,a}(X_t)$. It rules out i.i.d.-style cross-fitting schemes that, even indirectly, use contemporaneous or future outcomes when constructing the nuisance for time t. A simple way to enforce predictability is forward cross-fitting (Definition 4.3): partition the time axis into blocks $I_1, \ldots, I_K$, fit each nuisance model once per block using only data from $\cup_{j<k} I_j$, and hold the fitted model fixed while scoring units in I_k. To make this requirement auditable, an implementation should persist (i) the training indices used for each fit, (ii) fold assignments if internal cross-validation is used, and (iii) all learner randomness.

Lemma 4.6 (A sufficient operational condition for predictability)**.** *Under forward cross-fitting (Definition 4.3), if for each block I_k the analyst fits $(\widehat{m}_0^{(-k)}, \widehat{m}_1^{(-k)})$ using only data from blocks $I_1, \ldots, I_{k-1}$ and then reuses these fitted objects unchanged for all $t \in I_k$, then Assumption 4.5 holds.*

Proof. For $t \in I_k$, the fitted objects depend only on $\sigma(Z_s : s \in I_1 \cup \cdots \cup I_{k-1}) \subseteq \mathcal{F}_{t-1}$ and are therefore $\mathcal{F}_{t-1}$-measurable. □

Remark 4.7 (Connection to A2IPW terminology). Our $\widehat{\phi}_t$ is the same AIPW/DR pseudo-outcome used in the adaptive-experiment literature under the name *A2IPW* (adaptive AIPW). At time t the score is computed using outcome regressions fitted on past data only, together with the logged executed propensity used to randomize A_t (Kato et al., 2020; Cook et al., 2024). The present paper emphasizes that this predictability requirement is not merely a convenience: it is the condition that yields an exact martingale difference sequence and enables fixed-horizon Studentized inference without any stabilization of propensities or conditional variances.

4.3 The Estimator

Let $\mathcal{T} \subseteq \{1, \ldots, n\}$ denote the set of indices for which the nuisance used at time t is predictable (i.e., $\widehat{m}_{t-1}$ is $\mathcal{F}_{t-1}$-measurable). Under forward cross-fitting with a burn-in block I_1, one typically takes $\mathcal{T} := \{1, \ldots, n\} \setminus I_1$. Let $n_{\text{eff}} := \sum_{t=1}^{n} \mathbb{1}\{t \in \mathcal{T}\} = |\mathcal{T}|$. Throughout the asymptotic theory we treat the scored index set $\mathcal{T}$ as deterministic, as is the case under the forward-block construction in Definition 4.3.

Assumption 4.8 (Deterministic scored set)**.** The scored index set $\mathcal{T} \subseteq \{1, \ldots, n\}$ is fixed prior to data collection (equivalently, $T_t := \mathbb{1}\{t \in \mathcal{T}\}$ is nonrandom for each t).

Remark 4.9 (Predictable scored sets). If $T_t := \mathbb{1}\{t \in \mathcal{T}\}$ is allowed to be $\mathcal{F}_{t-1}$-measurable, the same proof strategy applies to $\tilde{\xi}_t := T_t(\widehat{\phi}_t - \theta_0)$. In this case, interpret $n_{\text{eff}} := \sum_{t=1}^{n} T_t$ and define $Q_{\mathcal{T}}$ and $V_{\mathcal{T}}^2$ using $\tilde{\xi}_t$. We omit further details of the predictable-$\mathcal{T}$ extension to keep the note focused.

Estimator and studentizing factor. The cross-fitted pseudo-outcome is

$$\widehat{\phi}_t := \phi_t(\widehat{m}_{t-1}, \pi_t), \qquad \widehat{\theta} := \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{T}} \widehat{\phi}_t. \tag{4.3}$$

where $\widehat{m}_{t-1}$ is the predictable nuisance estimate.

$$\widehat{V} := \frac{1}{n_{\text{eff}} - 1} \sum_{t \in \mathcal{T}} (\widehat{\phi}_t - \widehat{\theta})^2, \qquad \text{defined for } n_{\text{eff}} \geq 2. \tag{4.4}$$

Remark 4.10 (The studentizer as realized quadratic variation). Let $n_{\text{eff}} := |\mathcal{T}|$ and recall $\widehat{V}$ in (4.4) and $Q_{\mathcal{T}}$ in (5.3). A direct calculation gives

$$(n_{\text{eff}} - 1)\widehat{V} = Q_{\mathcal{T}} - n_{\text{eff}}(\widehat{\theta} - \theta_0)^2. \tag{4.5}$$

Thus $\widehat{V}$ is essentially the realized quadratic variation $Q_{\mathcal{T}}$ up to the negligible centering term $n_{\text{eff}}(\widehat{\theta} - \theta_0)^2/(n_{\text{eff}} - 1)$. Proposition 4.11 records a minimal condition under which $(n_{\text{eff}} - 1)\widehat{V}/Q_{\mathcal{T}} \to_p 1$.

Proposition 4.11 (Feasibility of the studentizer)**.** *Let $\xi_t := \widehat{\phi}_t - \theta_0$ and $S_{\mathcal{T}} := \sum_{t \in \mathcal{T}} \xi_t$, so that $\widehat{\theta} - \theta_0 = S_{\mathcal{T}}/n_{\text{eff}}$ and $Q_{\mathcal{T}} = \sum_{t \in \mathcal{T}} \xi_t^2$. If $n_{\text{eff}} \to \infty$ and $S_{\mathcal{T}}/\sqrt{Q_{\mathcal{T}}} = O_p(1)$, then*

$$\frac{(n_{\text{eff}} - 1)\widehat{V}}{Q_{\mathcal{T}}} \to_p 1.$$

In particular, the condition $S_{\mathcal{T}}/\sqrt{Q_{\mathcal{T}}} = O_p(1)$ holds whenever $S_{\mathcal{T}}/\sqrt{Q_{\mathcal{T}}} \Rightarrow \mathcal{N}(0, 1)$.

Proof. By (4.5),

$$\frac{(n_{\text{eff}} - 1)\widehat{V}}{Q_{\mathcal{T}}} = 1 - \frac{1}{n_{\text{eff}}} \left(\frac{S_{\mathcal{T}}}{\sqrt{Q_{\mathcal{T}}}} \right)^2,$$

which converges to 1 in probability under the stated conditions. □

Score sum notation. Define the centered increments $\xi_t := \widehat{\phi}_t - \theta_0$ and the score sum $S_{\mathcal{T}} := \sum_{t \in \mathcal{T}} \xi_t = n_{\text{eff}}(\widehat{\theta} - \theta_0)$.

$$\widehat{\text{SE}} := \sqrt{\widehat{V}/n_{\text{eff}}}. \tag{4.6}$$

$$\text{CI}_{1-\alpha} = \left[\widehat{\theta} \pm z_{1-\alpha/2}\, \widehat{\text{SE}} \right]. \tag{4.7}$$

Remark 4.12 (Finite-n reporting convention). For small n_{eff}, it is common as a pragmatic finite-sample convention to also report a t-critical variant using $t_{0.975, n_{\text{eff}}-1}$, namely $CI_t := \widehat{\theta} \pm t_{0.975, n_{\text{eff}}-1} \sqrt{\widehat{V}/n_{\text{eff}}}$, where $\widehat{V}$ is the sample variance in (4.4). This is a reporting convention; we do not claim finite-sample Student-t validity, and it is not covered by Theorem 5.14.

Algorithm 1 summarizes a concrete fixed-horizon implementation.

5 Main Theoretical Results

Remark 5.1 (Triangular-array and uniformity convention). All objects may depend on the horizon n (e.g. forward blocks and nuisance fits). We suppress the index n and assume overlap and moment constants are uniform over n.

This section isolates the logic behind predictable AIPW/DML in adaptive experiments. First, under logged executed propensities and predictable nuisance fits, the centered score is an exact martingale difference,

Algorithm 1 Auditable predictable AIPW/DML at a prespecified horizon

Require: Logged stream $\{(X_t, A_t, Y_t, \pi_t)\}_{t=1}^n$ with executed propensities; deterministic blocks $I_1, \ldots, I_K$; learner $\mathcal{L}$; level α.

Set scored set $\mathcal{T} := \bigcup_{k=2}^K I_k$ (burn-in I_1 excluded); if an arm has too few observations to fit a learner in early blocks, extend the burn-in or use a simple default model until both arms are observed.

for $k = 2$ to K **do**

  Fit $\widehat{m}_{k-1,a} \leftarrow \mathcal{L}\big(\{(X_s, Y_s) : s \in \bigcup_{j<k} I_j,\ A_s = a\}\big)$ for $a \in \{0, 1\}$.

  Persist (audit log) the training index set $\bigcup_{j<k} I_j$ and all learner randomness (seed, folds, tuning decisions).

  for each $t \in I_k$ **do**

    Compute $\widehat{\phi}_t \leftarrow \phi_t(\widehat{m}_{k-1}, \pi_t)$ as in (4.1).

Compute $\widehat{\theta}$ and $\widehat{V}$ from (4.3)–(4.4) over $\mathcal{T}$.

Output $\mathrm{CI}_{1-\alpha}$ in (4.7).

yielding finite-sample conditional unbiasedness (Proposition 5.2 and Lemma 5.3). Second, fixed-horizon inference follows from a self-normalized martingale CLT applied to the martingale sum, with feasibility provided by a sample-variance approximation (Theorem 5.14). Table 10 in Appendix E maps each assumption to each statement, and Appendix A states the martingale limit theorem invoked in the proofs.

5.1 The Martingale Structure

The central insight of our analysis is that forward cross-fitting induces a martingale difference structure. We first establish an identification result for the pseudo-outcome.

Proposition 5.2 (Identification, including predictable random nuisances)**.** *Let $m_{t-1} = (m_{t-1,0}, m_{t-1,1})$ be any (possibly random) pair of regression functions that is $\mathcal{F}_{t-1}$-measurable. Under Assumptions 3.6, 3.2, and 3.7, for each t,*

$$\mathbb{E}\big[\phi_t(m_{t-1}) \mid \mathcal{G}_t\big] = \tau(X_t). \tag{5.1}$$

Consequently, $\mathbb{E}[\phi_t(m_{t-1})] = \theta_0$.

Proof. Fix t and condition on $\mathcal{G}_t = \sigma(\mathcal{F}_{t-1}, X_t, \pi_t)$. Since m_{t-1} is $\mathcal{F}_{t-1}$-measurable and $\mathcal{F}_{t-1} \subseteq \mathcal{G}_t$, the function m_{t-1} is fixed under this conditioning.

By sequential randomization (Assumption 3.7),

$$\mathbb{E}[Y_t \mid \mathcal{G}_t, A_t = 1] = \mathbb{E}[Y_t(1) \mid \mathcal{G}_t], \qquad \mathbb{E}[Y_t \mid \mathcal{G}_t, A_t = 0] = \mathbb{E}[Y_t(0) \mid \mathcal{G}_t].$$

By i.i.d. arrivals (Assumption 3.6) and the exogeneity condition in Definition 3.1, $(Y_t(0), Y_t(1)) \perp\!\!\!\perp (\mathcal{F}_{t-1}, \pi_t) \mid X_t$ and thus $\mathbb{E}[Y_t(a) \mid X_t, \mathcal{F}_{t-1}, \pi_t] = \mathbb{E}[Y_t(a) \mid X_t] = m_a^\star(X_t)$ for $a \in \{0, 1\}$.

Using $\mathbb{E}[A_t \mid \mathcal{G}_t] = \pi_t$ and the tower property,

$$\begin{aligned}\mathbb{E}\left[\frac{A_t(Y_t - m_{t-1,1}(X_t))}{\pi_t}\,\middle|\,\mathcal{G}_t\right] &= \mathbb{E}\left[\frac{A_t}{\pi_t}\,\mathbb{E}[Y_t - m_{t-1,1}(X_t) \mid \mathcal{G}_t, A_t]\,\middle|\,\mathcal{G}_t\right] \\ &= \mathbb{E}\left[\frac{A_t}{\pi_t}\,(m_1^\star(X_t) - m_{t-1,1}(X_t))\,\middle|\,\mathcal{G}_t\right] \\ &= m_1^\star(X_t) - m_{t-1,1}(X_t),\end{aligned}$$

and similarly

$$\mathbb{E}\left[\frac{(1-A_t)(Y_t - m_{t-1,0}(X_t))}{1-\pi_t}\,\middle|\,\mathcal{G}_t\right] = m_0^\star(X_t) - m_{t-1,0}(X_t).$$

Plugging into the definition of $\phi_t(m_{t-1})$ in (4.1) yields

$$\begin{aligned}\mathbb{E}[\phi_t(m_{t-1}) \mid \mathcal{G}_t] &= (m_{t-1,1}(X_t) - m_{t-1,0}(X_t)) + (m_1^\star(X_t) - m_{t-1,1}(X_t)) \\ &\quad - (m_0^\star(X_t) - m_{t-1,0}(X_t)) \\ &= m_1^\star(X_t) - m_0^\star(X_t) = \tau(X_t),\end{aligned}$$

which proves (5.1). Taking unconditional expectations gives $\mathbb{E}[\phi_t(m_{t-1})] = \mathbb{E}[\tau(X_t)] = \theta_0$. □

The key consequence is that the pseudo-outcome, when evaluated with any predictable nuisance estimates, remains conditionally unbiased.

Lemma 5.3 (Martingale difference structure)**.** *Suppose Assumptions 3.6, 3.2, 3.7, 3.9, and 3.10 hold. Let $\{\widehat{m}_{t-1}\}$ satisfy Assumption 4.5. For each scored index $t \in \mathcal{T}$, define $\widehat{\phi}_t := \phi_t(\widehat{m}_{t-1})$ and $\xi_t := \widehat{\phi}_t - \theta_0$. Then*

$$\mathbb{E}[\widehat{\phi}_t \mid \mathcal{F}_{t-1}] = \theta_0, \tag{5.2}$$

and $\{\xi_t, \mathcal{F}_t\}$ is a martingale difference sequence over the scored indices.

Proof. (A finite first moment suffices here; this follows from Assumption 3.9. Higher-moment bounds used later are provided by Lemma 5.6.) Using iterated expectations and Proposition 5.2,

$$\mathbb{E}[\phi_t(\widehat{m}_{t-1}) \mid \mathcal{F}_{t-1}] = \mathbb{E}\big[\mathbb{E}[\phi_t(\widehat{m}_{t-1}) \mid \mathcal{G}_t] \mid \mathcal{F}_{t-1}\big] = \mathbb{E}[\tau(X_t) \mid \mathcal{F}_{t-1}].$$

By Assumption 3.6, X_t is independent of $\mathcal{F}_{t-1}$, so $\mathbb{E}[\tau(X_t) \mid \mathcal{F}_{t-1}] = \mathbb{E}[\tau(X)] = \theta_0$. □

Corollary 5.4 (Finite-sample unbiasedness)**.** *Under Assumptions 3.6, 3.2, 3.7, 4.8, 3.9, 3.10, and predictable nuisances as in Assumption 4.5, we have $\mathbb{E}[\widehat{\theta}] = \theta_0$.*

Proof. By linearity and Lemma 5.3,

$$\mathbb{E}[\widehat{\theta}] = \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{T}} \mathbb{E}[\phi_t(\widehat{m}_{t-1})] = \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{T}} \theta_0 = \theta_0.$$

□

Remark 5.5 (Interpretation). Lemma 5.3 is the conceptual replacement for fold-wise independence in i.i.d. DML. Rather than requiring approximate independence between nuisance estimation and score evaluation, we exploit exact conditional unbiasedness given the past. This martingale structure holds for any quality of nuisance estimates, provided they are predictable.

Lemma 5.6 (Moment bounds)**.** *Suppose Assumptions 3.9, 3.10, and 3.8 hold. Then there exists a constant $C < \infty$ such that*

$$\sup_{n\geq 1}\sup_{t\in\mathcal{T}} \mathbb{E}\big[|\widehat{\phi}_t|^4\big] \leq C.$$

Proof. Fix $t \in \mathcal{T}$. By overlap, $\pi_t \in [\varepsilon, 1-\varepsilon]$ almost surely, so the inverse-propensity weights are uniformly bounded on scored indices. Applying $(a+b+c+d)^4 \leq 4^3(a^4+b^4+c^4+d^4)$ to the score formula (4.1) (and hence to (4.3)) and using Assumptions 3.9 and 3.10 yields the claimed uniform bound. □

Remark 5.7 (Weaker than i.i.d.). Lemma 5.3 uses Assumption 3.6 only through the implication $\mathbb{E}[\tau(X_t) \mid \mathcal{F}_{t-1}] = \theta_0$. Thus the MDS argument extends to any arrival process satisfying this mean-stationarity condition.

5.2 The Variance Structure

Before stating the CLT, we analyze the variance structure. Define the realized and predictable quadratic variations:

$$Q_{\mathcal{T}} := \sum_{t\in\mathcal{T}} \xi_t^2, \qquad V_{\mathcal{T}}^2 := \sum_{t\in\mathcal{T}} \mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}]. \tag{5.3}$$

Proposition 5.8 (Conditional second-moment decomposition and oracle benchmark)**.** *Suppose Assumptions 3.6, 3.2, and 3.7 hold. Let $\sigma_a^2(X_t) := \mathrm{Var}(Y_t(a) \mid X_t)$ and define the regression bias terms $b_a(x) := m_a(x) - m_a^\star(x)$. Then for each $t \in \mathcal{T}$,*

$$\mathbb{E}[(\phi_t(m)-\theta_0)^2 \mid \mathcal{G}_t] = (\tau(X_t)-\theta_0)^2 + \frac{\sigma_1^2(X_t)}{\pi_t} + \frac{\sigma_0^2(X_t)}{1-\pi_t} + \pi_t(1-\pi_t)\left(\frac{b_1(X_t)}{\pi_t} + \frac{b_0(X_t)}{1-\pi_t}\right)^2. \tag{5.4}$$

Moreover, since $\mathbb{E}[\phi_t(m) \mid \mathcal{G}_t] = \tau(X_t)$ for any m that is $\mathcal{F}_{t-1}$-measurable, the last three terms in (5.4) equal $\mathrm{Var}(\phi_t(m) \mid \mathcal{G}_t)$. In particular, relative to the oracle score $\phi_t(m^\star)$ (for which $b_0 = b_1 \equiv 0$), using a misspecified m adds a nonnegative augmentation term to the conditional second moment and conditional variance.

Proof. Define $\varepsilon_{t,a} := Y_t(a) - m_a^\star(X_t)$, so that $\mathbb{E}[\varepsilon_{t,a} \mid X_t] = 0$ and $\mathbb{E}[\varepsilon_{t,a}^2 \mid X_t] = \sigma_a^2(X_t)$. A direct expansion using sequential randomization yields

$$\phi_t(m) - \tau(X_t) = \frac{A_t}{\pi_t}\varepsilon_{t,1} - \frac{1-A_t}{1-\pi_t}\varepsilon_{t,0} - \frac{A_t-\pi_t}{\pi_t}b_1(X_t) - \frac{A_t-\pi_t}{1-\pi_t}b_0(X_t),$$

from which (5.4) follows by taking conditional second moments given $\mathcal{G}_t$ and adding $(\tau(X_t)-\theta_0)^2$. Finally, since $\mathbb{E}[\phi_t(m) \mid \mathcal{G}_t] = \tau(X_t)$ for any m that is $\mathcal{F}_{t-1}$-measurable (in particular, any predictable learner), the last three terms in (5.4) are all centered at the oracle benchmark $\tau(X_t)$, and the proof is complete. □

Remark 5.9 (Oracle case and variance inflation are immediate). Setting $m = m^\star$ in Proposition 5.8 removes the nonnegative augmentation term and yields the oracle conditional second moment. The factors $1/\pi_t$ and

$1/(1-\pi_t)$ make explicit how extreme propensities inflate uncertainty, motivating overlap enforcement at the design stage.

Remark 5.10 (Why learn m without rates?). Theorem 5.14 yields valid prespecified-horizon coverage under predictability and logged propensities without requiring that $\widehat{m}_{t-1,a}$ converges to $m_a^\star$ at any particular rate. Learning m is therefore not needed for validity; it is needed for precision. Proposition 5.8 shows that the oracle score $\phi_t(m^\star)$ removes a nonnegative augmentation term from the conditional second moment, providing a natural benchmark for how nuisance quality affects the size of the studentizer. The oracle regression $m^\star$ minimizes the conditional variance within the AIPW score family (Proposition 5.8), and the augmentation term shrinks as m approaches $m^\star$; this does not imply monotone tightening relative to an arbitrary baseline across learning iterations. There is no monotonic guarantee relative to an arbitrary baseline. A poorly chosen nuisance can increase or decrease variance compared to another misspecified choice.

5.3 Studentized Asymptotic Normality

We now state our main inferential result. The key ingredient is that the realized quadratic variation $Q_{\mathcal{T}}$ and its predictable counterpart $V_{\mathcal{T}}^2$ are asymptotically equivalent under mild conditions, enabling studentization with the feasible sample variance $\widehat{V}$.

Roadmap. Lemma 5.3 identifies the scored sum $S_{\mathcal{T}}$ as a martingale sum with predictable quadratic variation $V_{\mathcal{T}}^2$ and realized quadratic variation $Q_{\mathcal{T}}$; Assumption 4.8 ensures that the scored indicator $\mathbb{1}\{t \in \mathcal{T}\}$ is deterministic (hence predictable). To apply the self-normalized martingale CLT in Theorem A.1, we verify: (i) $V_{\mathcal{T}}^2 \to \infty$ (Assumption 5.11); (ii) $Q_{\mathcal{T}}/V_{\mathcal{T}}^2 \to_p 1$ via a standard martingale variance-ratio argument, using the uniform fourth-moment bound from Lemma 5.6; and (iii) a Lyapunov condition, again by Lemma 5.6 together with Assumption 5.11. Finally, Proposition 4.11 justifies replacing $Q_{\mathcal{T}}$ by $(n_{\text{eff}}-1)\widehat{V}$ in the Studentized statistic.

Assumption 5.11 (Variance growth / nondegeneracy)**.** As $n \to \infty$, $n_{\text{eff}} := |\mathcal{T}| \to \infty$ and there exists $v_- > 0$ such that

$$\mathbb{P}\left(V_{\mathcal{T}}^2 \geq v_- n_{\text{eff}}\right) \to 1.$$

This is a nondegeneracy lower bound ensuring the predictable quadratic variation grows at least linearly in n_{eff}; it does not impose variance stabilization.

A simple primitive sufficient condition for Assumption 5.11 is that outcome noise is nondegenerate and overlap holds under the standing causal/sequential assumptions of Section 3. The following corollary records a sufficient condition; see Appendix C.3 for the proof.

Corollary 5.12 (A sufficient condition for variance growth)**.** *Suppose Assumptions 3.6, 3.2, 3.7, 3.8, and C.6 hold, where Assumption C.6 (Appendix) is the nondegeneracy condition* $\mathbb{E}[\sigma_1^2(X) + \sigma_0^2(X)] \geq \underline{\sigma}^2 > 0$. *Then Assumption 5.11 holds.*

Remark 5.13 (Nondegeneracy and numerical guardrails). Assumption 5.11 implies the quadratic variation grows, so $\widehat{V}$ is bounded away from zero with high probability asymptotically. In finite samples, if $\widehat{V} = 0$ (e.g. constant outcomes), inference is uninformative; report this as a design/measurement degeneracy.

Theorem 5.14 (Studentized fixed-horizon inference without variance stabilization)**.** *Suppose Assumptions 3.6, 3.2, 3.7, 3.8, 4.5, 4.8, 3.9, 3.10, and 5.11 hold. Let* $n_{\text{eff}} := |\mathcal{T}|$ *and recall* $S_{\mathcal{T}} = \sum_{t\in\mathcal{T}}(\widehat{\phi}_t - \theta_0)$ *and* $Q_{\mathcal{T}} = \sum_{t\in\mathcal{T}}(\widehat{\phi}_t - \theta_0)^2$*. Then*

$$\frac{S_{\mathcal{T}}}{\sqrt{Q_{\mathcal{T}}}} \Rightarrow \mathcal{N}(0,1), \tag{5.5}$$

and the practical Studentized statistic satisfies

$$\frac{\sqrt{n_{\text{eff}}}(\widehat{\theta} - \theta_0)}{\widehat{V}^{1/2}} \Rightarrow \mathcal{N}(0,1), \tag{5.6}$$

where $\widehat{V}$ *is defined in* (4.4)*. Consequently, the Wald interval* (4.7) *has asymptotically correct coverage at the prespecified horizon, without requiring variance stabilization (i.e.,* $V_{\mathcal{T}}^2/n_{\text{eff}} \to V$ *deterministic; Assumption C.1).*

Proof. Let $\xi_t := \mathbb{1}\{t \in \mathcal{T}\}(\widehat{\phi}_t - \theta_0)$ and $S_{\mathcal{T}} := \sum_{t\in\mathcal{T}}\xi_t$. Under Assumptions 3.6, 3.2, 3.7, 4.5, and 4.8, Lemma 5.3 implies that $(\xi_t, \mathcal{F}_t)$ is a martingale difference array. Assumption 5.11 gives $V_{\mathcal{T}}^2 := \sum_{t\in\mathcal{T}}\mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}] \to \infty$. Lemma 5.6 (which uses Assumptions 3.9, 3.10, and 3.8) provides a uniform fourth-moment bound for the scored increments; together with Assumption 5.11, a standard martingale variance-ratio calculation (see Appendix B) verifies the variance-ratio and Lyapunov conditions required by Theorem A.1. Therefore Theorem A.1 yields $S_{\mathcal{T}}/\sqrt{Q_{\mathcal{T}}} \Rightarrow \mathcal{N}(0,1)$, i.e., (5.5). Finally, Proposition 4.11 implies $(n_{\text{eff}} - 1)\widehat{V}/Q_{\mathcal{T}} \to_p 1$, and Slutsky's theorem yields (5.6). □

Remark 5.15 (Validity vs. precision and consistency). The studentized CLT in Theorem 5.14 does not require $\widehat{m}_{t-1}$ to converge to $m^\star$ for validity. Predictability (Assumption 4.5) together with overlap and moment/stability conditions (Assumptions 3.8–3.10) and variance growth (Assumption 5.11) are sufficient. Nuisance learning is nonetheless valuable: by Proposition 5.8 the oracle regression $m^\star$ minimizes the conditional variance within the AIPW score family (Proposition 5.8), and the augmentation term shrinks as m approaches $m^\star$; this does not imply monotone tightening relative to an arbitrary baseline across learning iterations. In particular, $\widehat{\theta}$ is consistent whenever $n_{\text{eff}} \to \infty$, and $\widehat{V}$ is a feasible studentizer even when $V_{\mathcal{T}}^2/n_{\text{eff}}$ does not stabilize.

Remark 5.16 (Relation to stabilization-based and anytime-valid approaches). Some fixed-horizon CLTs in the adaptive-experiment literature normalize by a deterministic asymptotic variance, imposing stabilization/design-stability conditions ensuring that an average conditional variance (or, equivalently, $V_{\mathcal{T}}^2/n_{\text{eff}}$) converges to a deterministic limit (Hadad et al., 2021; Zhan et al., 2021; Kato et al., 2021; Cook et al., 2024; Sengupta et al., 2025; Zenati et al., 2025). Such assumptions are appropriate when the deployed policy stabilizes or when one explicitly engineers stability.

Theorem 5.14 takes a different route. We normalize by the realized quadratic variation, so validity does not require a deterministic variance limit. This directly accommodates regimes where propensities oscillate, converge to random limits, or otherwise fail to stabilize, while still delivering a conventional fixed-horizon Wald interval. Finally, Theorem 5.14 is a prespecified-horizon result. If the experiment is continuously monitored and potentially stopped early based on the data, time-uniform methods such as confidence sequences and time-uniform CLTs are needed (Howard et al., 2021; Waudby-Smith and Ramdas, 2024; Waudby-Smith et al., 2024). Because our score increments form an exact MDS, the setup is compatible

with standard time-uniform martingale methods, but we do not develop anytime-valid inference here.

Appendix C collects three extensions not needed for the main methods-note claim: (i) a non-studentized CLT under variance stabilization; (ii) oracle equivalence under L^2 nuisance consistency; and (iii) a primitive sufficient condition for variance growth.

6 Simulation Study

We compare fixed-horizon 95% Wald intervals of the form (4.7) under different normalizations and nuisance-fitting regimes, depending on the design:

(CI1) **SN (self-normalized):** the proposed studentized Wald interval (4.7).

(CI2) **Fixed-V (stabilization-style baseline, Designs A–B):** a Wald interval that replaces the (potentially random) long-run variance by a fixed constant V_{fix},

$$\widehat{\theta} \pm z_{1-\alpha/2}\sqrt{V_{\text{fix}}/n_{\text{eff}}},$$

thereby behaving as if Assumption C.1 held with deterministic limit. We specify V_{fix} explicitly in each design.

(CI3) **Leaky baselines (Designs C1 and D):** nuisance fits that violate predictability (Assumption 4.5) by using contemporaneous or future outcomes when scoring unit t. When computed, the score is still (4.1) and the CI is (4.7), but the fixed-horizon guarantee of Theorem 5.14 does not apply.

(CI4) **Regime-aware Fixed-V (Design A only):** an infeasible regime-aware fixed-variance benchmark that plugs in the correct regime-specific V_{fix} based on the burn-in sign.

For each design we report: (i) two-sided 95% coverage; (ii) average CI length; and (iii) where informative, mean bias and/or null rejection rates. Monte Carlo standard errors for coverage are $(\sqrt{\hat{p}(1-\hat{p})/R})$. Tables report n and n_{eff} explicitly. Unless noted, we use $R = 1000$ replications and $n \in \{250, 500, 1000, 2000, 5000\}$.

6.1 Design A: random long-run variance (stabilization fails)

During burn-in, $\pi_t = 0.5$ for $t \le n_0 = 50$, and we compute the burn-in estimate

$$\widehat{\tau}_{\text{burn}} := \frac{1}{n_0}\sum_{t=1}^{n_0}\left(\frac{A_t Y_t}{\pi_t} - \frac{(1-A_t)Y_t}{1-\pi_t}\right)$$

using only burn-in units. After burn-in, if $\widehat{\tau}_{\text{burn}} \ge 0$ then $\pi_t = 0.8$, else $\pi_t = 0.2$. Outcomes: $Y(0) \sim \mathcal{N}(0,1)$ and $Y(1) = \varepsilon_1$ with $\varepsilon_1 \sim \mathcal{N}(0,9)$. There are no covariates; with $m_0 \equiv m_1 \equiv 0$, the AIPW score (4.1) reduces to the usual inverse-propensity weighted (IPW) score.

On $\mathcal{T} = \{n_0+1, \dots, n\}$ we have $n_{\text{eff}} = n - n_0$ and π_t is constant after burn-in, so the oracle conditional variance converges to a *random* limit:

$$\frac{V_{\mathcal{T}}^2}{n_{\text{eff}}} \to \begin{cases} 9/0.8 + 1/0.2 = 16.25, & \widehat{\tau}_{\text{burn}} \ge 0, \\ 9/0.2 + 1/0.8 = 46.25, & \widehat{\tau}_{\text{burn}} < 0. \end{cases}$$

Thus Assumption C.1 fails (no deterministic variance limit). The SN CI remains valid under Theorem 5.14 because it self-normalizes by realized quadratic variation. SN uses (4.7). Fixed-V sets $V_{\text{fix}} := 31.25$ (the unconditional mean of the two variance limits above). Regime-aware Fixed-V plugs in $V_{\text{fix}} = 16.25$ or 46.25 depending on the realized burn-in sign.

Table 3: Design A (random long-run variance): coverage and length of 95% CIs.

n	n_{eff}	Method	Coverage	MCSE	Avg. length
250	200	Fixed-V	0.944	(0.007)	1.549
250	200	Regime-Fixed	0.950	(0.007)	1.515
250	200	SN	0.955	(0.007)	1.506
500	450	Fixed-V	0.939	(0.008)	1.033
500	450	Regime-Fixed	0.950	(0.007)	1.003
500	450	SN	0.957	(0.006)	1.002
1000	950	Fixed-V	0.938	(0.008)	0.711
1000	950	Regime-Fixed	0.948	(0.007)	0.697
1000	950	SN	0.948	(0.007)	0.696
2000	1950	Fixed-V	0.947	(0.007)	0.496
2000	1950	Regime-Fixed	0.941	(0.007)	0.477
2000	1950	SN	0.946	(0.007)	0.476
5000	4950	Fixed-V	0.941	(0.007)	0.311
5000	4950	Regime-Fixed	0.960	(0.006)	0.303
5000	4950	SN	0.959	(0.006)	0.303

Notes: $R = 1000$ replications; $n \in \{250, 500, 1000, 2000, 5000\}$; burn-in $n_0 = 50$ (so $n_{\text{eff}} = n - 50$). SN is the studentized interval (4.7). Fixed-V uses $V_{\text{fix}} = 31.25$, the unconditional mean of the regime-specific variance limits. Regime-aware Fixed-V is an infeasible oracle benchmark that plugs in the correct regime-specific variance term (i.e., $\sigma_1^2 = 9$ and $\sigma_0^2 = 1$, so the regime-specific limits are $9/0.8 + 1/0.2 = 16.25$ and $9/0.2 + 1/0.8 = 46.25$), and uses z-critical values.

Table 3 reports marginal coverage and average length for Design A as n increases. In this design, the studentized SN interval remains close to the nominal 95% level across horizons, consistent with Theorem 5.14. The fixed-V interval normalizes by a single deterministic target V that averages over the two post–burn-in propensity regimes; as a consequence, its *marginal* coverage can be close to nominal while still exhibiting regime-dependent over/under coverage (Table 4). The regime-aware fixed-V benchmark—which plugs in the correct regime-specific variance constant—illustrates that classical normalization works once one conditions on (and correctly accounts for) the realized variance regime; the SN interval achieves this conditioning automatically via studentization.

Design A illustrates the basic phenomenon targeted by Theorem 5.14: the variance proxy $V_{\mathcal{T}}^2/n_{\text{eff}}$ does not converge to a single deterministic constant because it depends on the realized post–burn-in propensity regime. Table 4 shows that a fixed-V normalization can be conservative when the realized regime has low variance ($\pi_t = 0.8$) and anti-conservative when the realized regime has high variance ($\pi_t = 0.2$). In contrast, the SN interval studentizes by the realized quadratic variation and maintains near-nominal conditional coverage in both regimes.

Method	Post burn-in π	$n = 250$	$n = 500$	$n = 1000$	$n = 2000$
SN	0.8	0.96	0.97	0.95	0.95
SN	0.2	0.95	0.95	0.94	0.94
Fixed-V	0.8	1.00	0.99	0.99	1.00
Fixed-V	0.2	0.89	0.89	0.89	0.89

Notes: Each row conditions on the realized post–burn-in propensity in Design A: if $\widehat{\tau}_{\text{burn}} \geq 0$ then $\pi_t = 0.8$ for $(t > n_0)$, and if $\widehat{\tau}_{\text{burn}} < 0$ then $\pi_t = 0.2$ for $(t > n_0)$.

Table 4: Conditional coverage by realized propensity regime (Design A)

6.2 Design B: stabilization benchmark

Reuse Design A outcomes with $\tau = 0$, $Y(0) \sim \mathcal{N}(0, 1)$ and $Y(1) = \varepsilon_1$ with $\varepsilon_1 \sim \mathcal{N}(0, 9)$; set $\pi_t \equiv 0.6$ for all $t \leq n$ (no burn-in, so $n_{\text{eff}} = n$). With no covariates, AIPW reduces to IPW, and Assumption C.1 holds with deterministic long-run variance

$$V_{\text{fix}} \;=\; 9/0.6 + 1/0.4 \;=\; 17.5,$$

so SN and Fixed-V coincide asymptotically.

Table 5: Design B (stabilization benchmark): coverage and length of 95% CIs.

n	n_{eff}	Method	Coverage	MCSE	Avg. length
250	250	Fixed-V	0.954	(0.007)	1.037
250	250	SN	0.952	(0.007)	1.038
500	500	Fixed-V	0.949	(0.007)	0.733
500	500	SN	0.953	(0.007)	0.733
1000	1000	Fixed-V	0.957	(0.006)	0.519
1000	1000	SN	0.955	(0.007)	0.518
2000	2000	Fixed-V	0.953	(0.007)	0.367
2000	2000	SN	0.953	(0.007)	0.366
5000	5000	Fixed-V	0.947	(0.007)	0.232
5000	5000	SN	0.947	(0.007)	0.232

Notes: $R = 1000$ replications; $n \in \{250, 500, 1000, 2000, 5000\}$; $n_{\text{eff}} = n$. SN is (4.7). Fixed-V uses $V_{\text{fix}} = 17.5$.

Table 5 shows that when the variance stabilizes, SN and Fixed-V give nearly identical coverage and length. Differences are within Monte Carlo error, consistent with the deterministic V_{fix}.

6.3 Design C1: leakage stress test (predictability)

Design C1 is a stress test for Assumption 4.5 and Remark 4.4. Covariates are $X_t \in \mathbb{R}^p$ with $p = 20$ and i.i.d. $X_t \sim \mathcal{N}(0, I_p)$. Potential outcomes follow

$$Y_t(0) = m_0(X_t) + \varepsilon_{t0}, \qquad Y_t(1) = m_0(X_t) + \tau(X_t) + \varepsilon_{t1},$$

where $m_0(x)$ is nonlinear and $\tau(x) = \tau_0 + \delta \sin(x_1)$ so the superpopulation ATE equals $\theta_0 = \mathbb{E}[\tau(X_t)] = \tau_0$. We report calibration at $\tau_0 = 0$ with $\delta = 0.2$; errors are independent t_{30} draws rescaled to unit variance (finite moments, but heavier tails than Gaussian).

To induce adaptive feedback, we split $\{1, \ldots, n\}$ into $K = 5$ contiguous blocks and use the first block as burn-in with $\pi_t \equiv 0.5$. On subsequent blocks, the executed propensity is

$$\pi_t = \varepsilon + (1 - 2\varepsilon)\mathrm{expit}\big(\lambda\, \widehat{\tau}_{t-1}(X_t)\big), \qquad \widehat{\tau}_{t-1}(x) = \widehat{m}_{t-1,1}(x) - \widehat{m}_{t-1,0}(x),$$

with $\varepsilon = 0.1$ and $\lambda = 2.5$. The *predictable* nuisance $\widehat{m}_{t-1,a}$ is fit by ridge regression on a low-dimensional feature map using only prior blocks and held fixed within the current block (forward fitting). The *leaky* baseline fits a richer ridge model once on the full sample (uses future outcomes), violating predictability.

Table 6: Design C1 (predictability / leakage stress test, $\tau_0 = 0$): coverage, length, and null rejection.

n	n_{eff}	Method	Coverage	MCSE	Avg. length	Reject rate
250	200	SN-AIPW-Predictable	0.952	(0.007)	0.653	0.048
250	200	SN-AIPW-LeakyFull	0.821	(0.012)	1.864	0.179
250	200	SN-IPW	0.939	(0.008)	0.766	0.061
500	400	SN-AIPW-Predictable	0.953	(0.007)	0.429	0.047
500	400	SN-AIPW-LeakyFull	0.887	(0.010)	0.398	0.113
500	400	SN-IPW	0.958	(0.006)	0.525	0.042
1000	800	SN-AIPW-Predictable	0.948	(0.007)	0.290	0.052
1000	800	SN-AIPW-LeakyFull	0.928	(0.008)	0.274	0.072
1000	800	SN-IPW	0.947	(0.007)	0.360	0.053
2000	1600	SN-AIPW-Predictable	0.957	(0.006)	0.202	0.043
2000	1600	SN-AIPW-LeakyFull	0.944	(0.007)	0.195	0.056
2000	1600	SN-IPW	0.962	(0.006)	0.251	0.038
5000	4000	SN-AIPW-Predictable	0.944	(0.007)	0.126	0.056
5000	4000	SN-AIPW-LeakyFull	0.941	(0.007)	0.124	0.059
5000	4000	SN-IPW	0.951	(0.007)	0.157	0.049

Notes: $R = 1000$ replications; $K = 5$ contiguous blocks with the first block omitted from scoring (so $n_{\text{eff}} = 4n/5$). The executed propensity is $\pi_t = \varepsilon + (1 - 2\varepsilon)\mathrm{expit}(\lambda\, \widehat{\tau}_{t-1}(X_t))$ with $\varepsilon = 0.1$ and $\lambda = 2.5$. "SN-AIPW-Predictable" uses forward (past-only) ridge nuisances; "SN-AIPW-LeakyFull" fits a full-sample ridge nuisance on richer features and is not predictable. Reject rate is the two-sided rejection probability of $H_0 : \theta_0 = 0$ based on the corresponding 95% CI.

Table 6 makes the predictability requirement operational: the predictable SN-AIPW procedure remains near nominal, while the leaky full-sample nuisance substantially undercovers and over-rejects at smaller horizons, consistent with the warning in Remark 4.4.

6.4 Design C2: nuisance quality as a precision knob

Design C2 isolates how nuisance quality affects precision without confounding from adaptivity. Covariates $X_t \in \mathbb{R}^5$ follow a correlated normal with AR(1) correlation $\rho = 0.5$; treatment is assigned with constant propensity $\pi_t \equiv 0.5$. Potential outcomes are linear with homoskedastic noise:

$$Y_t(0) = X_{t1} + X_{t2} + \varepsilon_{t0}, \qquad Y_t(1) = X_{t1} + X_{t2} + \tau + \varepsilon_{t1}, \qquad \varepsilon_{t0}, \varepsilon_{t1} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1),$$

so the oracle regression removes the augmentation term in Proposition 5.8. We use $K = 10$ forward blocks (first block omitted) and report results at $n = 5000$ under $\tau = 0$.

Table 7: Design C2 (precision vs nuisance quality, $n = 5000$): coverage and relative length/variance vs oracle.

n	n_{eff}	Method	Coverage	MCSE	Avg. length	Rel. length	Rel. $\widehat{V}$
5000	4500	SN-AIPW-Oracle	0.955	(0.007)	0.117	1.000	1.000
5000	4500	SN-AIPW-WellSpec	0.951	(0.007)	0.117	1.002	1.004
5000	4500	SN-AIPW-Misspec	0.956	(0.006)	0.155	1.323	1.750
5000	4500	SN-IPW	0.937	(0.008)	0.234	1.999	3.997

Notes: $R = 1000$ replications; $\pi_t \equiv 0.5$; $K = 10$ forward blocks with the first block omitted from scoring (so $n_{\text{eff}} = 4500$). "WellSpec" fits arm-specific linear regression on all covariates; "Misspec" fits on a restricted covariate set; "SN-IPW" sets $m \equiv 0$. Relative quantities are computed with respect to the oracle procedure within the same design.

Table 7 shows the oracle benchmarking message: coverage remains near nominal across nuisance choices, while interval length (and $\widehat{V}$) inflates as the nuisance becomes more misspecified, consistent with Proposition 5.8 and the "validity vs precision" discussion following Theorem 5.14.

6.5 Design D: adaptive policy and logging integrity

Design D is a simple contextual-bandit-style adaptive experiment that highlights the design-time role of executed propensity logging (Assumption 3.2). Covariates are $X_t \in \mathbb{R}^{10}$ with correlated normal $X_t \sim \mathcal{N}(0, \Sigma)$ where $\Sigma_{ij} = \rho^{|i-j|}$ and $\rho = 0.3$. The baseline outcome, heterogeneous treatment effect, and heteroskedastic noise are

$$m_0(x) = 0.8x_1 + 0.5x_2^2 - 0.5\cos(x_3) + 0.25x_4, \qquad \tau(x) = 0.5x_1 + 0.5\sin(x_2) + 0.25\mathbb{1}\{x_3 > 0\} - 0.25x_4x_5,$$

$$\sigma(x) = 1 + 0.5|x_1|, \qquad Y_t(a) = m_a(X_t) + \varepsilon_{ta}, \;\; \varepsilon_{ta} \mid X_t \sim \mathcal{N}(0, \sigma(X_t)^2),$$

so $\theta_0 = \mathbb{E}[\tau(X_t)]$ is the ATE. We use burn-in $n_0 = 100$ with $\pi_t \equiv 0.5$ and then update in blocks of size 100: on each block we fit arm-specific linear regressions on past data only and choose the executed propensity by either an ε-greedy rule or a softmax rule, clipping to $[0.05, 0.95]$ to enforce overlap.

We compare: (i) **SN-AIPW** with predictable (past-only) nuisance fits; (ii) **Naive-iid-DML** using leaky 5-fold cross-fitting that ignores time ordering; (iii) **SN-IPW** with $m \equiv 0$; (iv) **SN-Oracle** using the true m_0, m_1; and (v) **SN-IPW-Assume0p5**, an analysis-time mis-logging baseline that computes the score using $\pi_t \equiv 0.5$ rather than the logged executed propensity π_t.

Tables 8–9 show two complementary messages: with the *logged executed* propensities and predictable nuisance fitting, the studentized intervals are close to nominal under both adaptive policies; in contrast, the mis-logged "Assume0p5" baseline is severely biased and has catastrophic undercoverage, illustrating why Assumption 3.2 is design-critical.

Table 8: Design D (adaptive assignment, ε-greedy policy): coverage, length, and bias of 95% CIs.

n	n_{eff}	Method	Coverage	MCSE	Avg. length	Bias
250	150	SN-Oracle	0.946	(0.007)	1.516	0.009
250	150	SN-AIPW	0.944	(0.007)	1.906	0.019
250	150	Naive-iid-DML	0.944	(0.007)	1.883	0.019
250	150	SN-IPW	0.934	(0.008)	2.083	0.003
250	150	SN-IPW-Assume0p5	0.433	(0.016)	1.296	0.703
500	400	SN-Oracle	0.950	(0.007)	0.940	-0.008
500	400	SN-AIPW	0.939	(0.008)	1.174	-0.012
500	400	Naive-iid-DML	0.943	(0.007)	1.166	-0.014
500	400	SN-IPW	0.951	(0.007)	1.304	-0.017
500	400	SN-IPW-Assume0p5	0.064	(0.008)	0.795	0.869
1000	900	SN-Oracle	0.948	(0.007)	0.629	0.003
1000	900	SN-AIPW	0.954	(0.007)	0.784	-0.006
1000	900	Naive-iid-DML	0.950	(0.007)	0.786	-0.006
1000	900	SN-IPW	0.947	(0.007)	0.864	-0.003
1000	900	SN-IPW-Assume0p5	0.005	(0.002)	0.528	0.943
2000	1900	SN-Oracle	0.953	(0.007)	0.433	0.001
2000	1900	SN-AIPW	0.955	(0.007)	0.543	0.001
2000	1900	Naive-iid-DML	0.948	(0.007)	0.548	0.001
2000	1900	SN-IPW	0.952	(0.007)	0.596	-0.003
2000	1900	SN-IPW-Assume0p5	0.001	(0.001)	0.363	0.979
5000	4900	SN-Oracle	0.958	(0.006)	0.271	-0.001
5000	4900	SN-AIPW	0.955	(0.007)	0.342	0.000
5000	4900	Naive-iid-DML	0.954	(0.007)	0.347	0.001
5000	4900	SN-IPW	0.955	(0.007)	0.372	-0.002
5000	4900	SN-IPW-Assume0p5	0.003	(0.002)	0.226	0.984

Notes: $R = 1000$ replications; burn-in $n_0 = 100$ so $n_{\text{eff}} = n - 100$; blocks of size 100. The executed propensity is chosen by an ε-greedy rule with $\varepsilon = 0.1$ and then clipped to $[0.05, 0.95]$. Bias is $\mathbb{E}[\widehat{\theta} - \theta_0]$ (Monte Carlo mean), where $\theta_0 = \mathbb{E}[\tau(X)]$ is approximated by Monte Carlo integration under the covariate distribution.

Table 9: Design D (adaptive assignment, softmax policy): coverage, length, and bias of 95% CIs.

n	n_{eff}	Method	Coverage	MCSE	Avg. length	Bias
250	150	SN-Oracle	0.953	(0.007)	1.509	-0.003
250	150	SN-AIPW	0.948	(0.007)	1.953	0.012
250	150	Naive-iid-DML	0.940	(0.008)	1.934	0.013
250	150	SN-IPW	0.916	(0.009)	2.115	0.002
250	150	SN-IPW-Assume0p5	0.395	(0.015)	1.298	0.731
500	400	SN-Oracle	0.960	(0.006)	0.937	0.004
500	400	SN-AIPW	0.958	(0.006)	1.255	0.002
500	400	Naive-iid-DML	0.959	(0.006)	1.250	0.006
500	400	SN-IPW	0.947	(0.007)	1.338	0.004
500	400	SN-IPW-Assume0p5	0.088	(0.009)	0.795	0.840
1000	900	SN-Oracle	0.952	(0.007)	0.637	-0.005
1000	900	SN-AIPW	0.958	(0.006)	0.881	-0.011
1000	900	Naive-iid-DML	0.950	(0.007)	0.896	-0.008
1000	900	SN-IPW	0.947	(0.007)	0.919	0.003
1000	900	SN-IPW-Assume0p5	0.009	(0.003)	0.530	0.930
2000	1900	SN-Oracle	0.957	(0.006)	0.452	0.007
2000	1900	SN-AIPW	0.951	(0.007)	0.636	0.001
2000	1900	Naive-iid-DML	0.954	(0.007)	0.654	0.001
2000	1900	SN-IPW	0.935	(0.008)	0.651	0.006
2000	1900	SN-IPW-Assume0p5	0.004	(0.002)	0.365	0.982
5000	4900	SN-Oracle	0.954	(0.007)	0.290	0.001
5000	4900	SN-AIPW	0.954	(0.007)	0.421	0.005
5000	4900	Naive-iid-DML	0.952	(0.007)	0.435	0.005
5000	4900	SN-IPW	0.944	(0.007)	0.417	0.009
5000	4900	SN-IPW-Assume0p5	0.001	(0.001)	0.227	1.015

Notes: Same design as Table 8, but the executed propensity uses a softmax rule $\pi_t = \text{expit}(\widehat{\tau}_{t-1}(X_t)/T)$ with temperature $T = 0.5$ and clipping to $[0.05, 0.95]$.

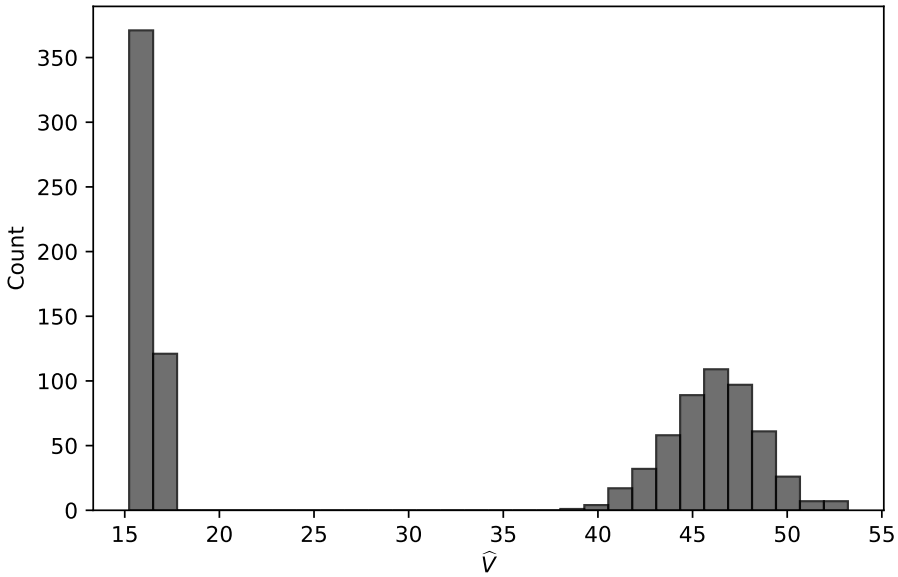

Figure 1: Design A: distribution across replications of the regime-dependent long-run variance proxy. In this design, $V_{\mathcal{T}}^2/n_{\text{eff}}$ converges to 16.25 or 46.25 depending on the burn-in sign, so there is no deterministic variance limit (Assumption C.1 fails). This motivates studentization by realized quadratic variation in the SN CI (4.7).

6.6 Variance-ratio variability

Figure 1 shows two distinct variance modes for $V_{\mathcal{T}}^2/n_{\text{eff}}$ induced by the post–burn-in propensity regime in Design A. This bimodality explains the regime-dependent length and coverage patterns in Table 4: intervals that plug in a single deterministic variance target are too short in the high-variance regime and too long in the low-variance regime, even if their marginal coverage can be closer to nominal by averaging over regimes. The SN interval uses the realized quadratic variation (equivalently $\widehat{V}$) within each replication and therefore adapts to the realized variance regime by construction.

7 Conclusion

We studied prespecified-horizon inference for the superpopulation ATE in sequentially adaptive experiments with logged propensities. Under the standard causal model (SUTVA/no interference and sequential randomization with known executed propensities), i.i.d. arrivals, executed overlap on a prespecified scored set, and predictable nuisance construction, the centered AIPW/DML pseudo-outcomes form a martingale difference sequence (Lemma 5.3). Studentizing by realized quadratic variation then yields a conventional fixed-horizon Wald approximation that remains valid even when the predictable variance path does not converge to a deterministic limit (Theorem 5.14). Valid coverage does not require nuisance consistency; rather, nuisance learning affects precision through the size of the realized quadratic variation. Proposition 5.8 provides an oracle benchmark: relative to the oracle score $\phi_t(m^\star)$, misspecification contributes an additional nonnegative term to the conditional second moment, and under L^2 convergence the feasible procedure is asymptotically oracle-equivalent (Theorem C.4). Finally, our "data contract" (Appendix F) isolates the pipeline conditions that are auditable from logs—executed propensity logging and predictable fitting—from

substantive assumptions that are not fully testable. The resulting confidence intervals are fixed-horizon only and should not be used under optional stopping; sequential monitoring requires time-uniform methods.

A Martingale Limit Theory

We use a self-normalized martingale CLT for (possibly triangular-array) martingale differences. Background and related results on martingale CLTs and self-normalization can be found in Hall and Heyde (1980), the monograph de la Peña et al. (2009), and the survey Shao and Wang (2013); see also modern bounds and concentration results for self-normalized martingales in Fan and Shao (2018); Fan et al. (2019); Bercu and Touati (2019). The statement below is a direct corollary of Hall and Heyde (1980, Theorem 3.10) combined with Slutsky; we record it in the form used in our proof of Theorem 5.14.

Theorem A.1 (Self-normalized martingale CLT (Lyapunov form))**.** *Let $(\xi_{n,t}, \mathcal{F}_{n,t})$ be a martingale difference array with partial sums $S_n = \sum_{t=1}^{k_n} \xi_{n,t}$. Define the predictable and realized quadratic variations*

$$V_n^2 := \sum_{t=1}^{k_n} \mathbb{E}[\xi_{n,t}^2 \mid \mathcal{F}_{n,t-1}] \qquad \textit{and} \qquad Q_n := \sum_{t=1}^{k_n} \xi_{n,t}^2.$$

Assume: (i) $V_n^2 \to_p \infty$; (ii) $\mathbb{P}(Q_n > 0) \to 1$ and $Q_n/V_n^2 \to_p 1$; and (iii) for some $\delta > 0$,

$$\frac{1}{V_n^{2+\delta}} \sum_{t=1}^{k_n} \mathbb{E}[|\xi_{n,t}|^{2+\delta}] \to 0.$$

Then $S_n/\sqrt{Q_n} \Rightarrow \mathcal{N}(0,1)$.

B Proofs

B.1 Proof of Theorem 5.14

Proof of Theorem 5.14. The proof proceeds in two steps.

Indexing convention. To match Theorem A.1 (stated for sums over $t = 1, \dots, n$), define the extended increments

$$\tilde{\xi}_t := T_t(\widehat{\phi}_t - \theta_0), \qquad T_t := \mathbb{1}\{t \in \mathcal{T}\}, \qquad t = 1, \dots, n.$$

Since $\mathcal{T}$ is deterministic, T_t is nonrandom and $\mathbb{E}[\tilde{\xi}_t \mid \mathcal{F}_{t-1}] = T_t \mathbb{E}[\widehat{\phi}_t - \theta_0 \mid \mathcal{F}_{t-1}] = 0$, so $\{\tilde{\xi}_t, \mathcal{F}_t\}$ is a martingale difference array. $\sum_{t=1}^n \tilde{\xi}_t = S_{\mathcal{T}}$, $\sum_{t=1}^n \tilde{\xi}_t^2 = Q_{\mathcal{T}}$, and $\sum_{t=1}^n \mathbb{E}[\tilde{\xi}_t^2 \mid \mathcal{F}_{t-1}] = V_{\mathcal{T}}^2$. For readability we drop the tilde notation below.

Step 1: Self-normalized CLT for $S_{\mathcal{T}}/\sqrt{Q_{\mathcal{T}}}$. Let $S_{\mathcal{T}} = \sum_{t \in \mathcal{T}} \xi_t = n_{\text{eff}}(\widehat{\theta} - \theta_0)$. We apply Theorem A.1 (stated in Appendix A). Assumption 5.11 implies $V_{\mathcal{T}}^2 \to \infty$ in probability.

To show $Q_{\mathcal{T}}/V_{\mathcal{T}}^2 \xrightarrow{p} 1$, define the martingale differences $\Delta_t := \xi_t^2 - \mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}]$ and the partial sums $M_k := \sum_{t=1}^k \Delta_t$ for $k = 1, \dots, n$. Then $\{M_k, \mathcal{F}_k\}$ is a martingale and $M_n = Q_{\mathcal{T}} - V_{\mathcal{T}}^2$. By Lemma 5.6, $\sup_t \mathbb{E}[\xi_t^4] < \infty$, and Jensen implies $\mathbb{E}[\mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}]^2] \le \mathbb{E}[\xi_t^4]$, hence $\sup_t \mathbb{E}[\Delta_t^2] \le 4 \sup_t \mathbb{E}[\xi_t^4] < \infty$. By orthogonality of martingale differences,

$$\mathbb{E}[M_n^2] = \sum_{t \in \mathcal{T}} \mathbb{E}[\Delta_t^2] = O(n_{\text{eff}}).$$

Let $\mathcal{E}_{\mathcal{T}} := \{V_{\mathcal{T}}^2 \geq v_- n_{\text{eff}}\}$, which satisfies $\mathbb{P}(\mathcal{E}_{\mathcal{T}}) \to 1$ by Assumption 5.11. On $\mathcal{E}_{\mathcal{T}}$,

$$\left| \frac{Q_{\mathcal{T}}}{V_{\mathcal{T}}^2} - 1 \right| \leq \frac{|M_n|}{v_- n_{\text{eff}}}.$$

Therefore, for any $\eta > 0$,

$$\mathbb{P}\left(\left| \frac{Q_{\mathcal{T}}}{V_{\mathcal{T}}^2} - 1 \right| > \eta \right) \leq \mathbb{P}(\mathcal{E}_{\mathcal{T}}^c) + \frac{\mathbb{E}[M_n^2]}{\eta^2 v_-^2 n_{\text{eff}}^2} \to 0,$$

so $Q_{\mathcal{T}}/V_{\mathcal{T}}^2 \xrightarrow{p} 1$.

For the Lyapunov condition with $\delta = 2$, Lemma 5.6 gives $\sup_{t \in \mathcal{T}} \mathbb{E}[|\widehat{\phi}_t|^4] < \infty$; since $\xi_t = T_t(\widehat{\phi}_t - \theta_0)$ and $T_t \in \{0, 1\}$, it follows that $\sup_{t \leq n} \mathbb{E}[|\xi_t|^4] < \infty$ for each n. Jensen then implies $\sup_{t \leq n} \mathbb{E}[|\xi_t|^{2+\delta}] < \infty$ for any $\delta \in (0, 2]$.

$$\frac{\sum_{t \in \mathcal{T}} \mathbb{E}|\xi_t|^4}{V_{\mathcal{T}}^4} \leq \frac{C n_{\text{eff}}}{(v_- n_{\text{eff}})^2} \to 0,$$

hence condition (iii) of Theorem A.1 holds. By Theorem A.1, $S_{\mathcal{T}}/\sqrt{Q_{\mathcal{T}}} \xrightarrow{d} \mathcal{N}(0, 1)$.

Step 2: Replace $Q_{\mathcal{T}}$ with $(n_{\text{eff}} - 1)\widehat{V}$. We have

$$\begin{aligned}
(n_{\text{eff}} - 1)\widehat{V} &= \sum_{t \in \mathcal{T}} (\widehat{\phi}_t - \widehat{\theta})^2 \\
&= \sum_{t \in \mathcal{T}} (\xi_t - (\widehat{\theta} - \theta_0))^2 \\
&= Q_{\mathcal{T}} - n_{\text{eff}}(\widehat{\theta} - \theta_0)^2.
\end{aligned}$$

Since $S_{\mathcal{T}}/\sqrt{Q_{\mathcal{T}}} = O_{\mathbb{P}}(1)$, we have $S_{\mathcal{T}}^2/Q_{\mathcal{T}} = O_{\mathbb{P}}(1)$, hence

$$\begin{aligned}
\frac{n_{\text{eff}}(\widehat{\theta} - \theta_0)^2}{Q_{\mathcal{T}}} &= \frac{S_{\mathcal{T}}^2}{n_{\text{eff}} Q_{\mathcal{T}}} \\
&= \frac{1}{n_{\text{eff}}} \cdot \frac{S_{\mathcal{T}}^2}{Q_{\mathcal{T}}} = o_{\mathbb{P}}(1).
\end{aligned}$$

Thus $(n_{\text{eff}} - 1)\widehat{V}/Q_{\mathcal{T}} \xrightarrow{p} 1$, and by Slutsky's lemma:

$$\frac{S_{\mathcal{T}}}{\sqrt{(n_{\text{eff}} - 1)\widehat{V}}} = \frac{S_{\mathcal{T}}}{\sqrt{Q_{\mathcal{T}}}} \cdot \sqrt{\frac{Q_{\mathcal{T}}}{(n_{\text{eff}} - 1)\widehat{V}}} \xrightarrow{d} \mathcal{N}(0, 1). \qquad \square$$

C Additional results

This appendix collects results on variance stabilization, oracle equivalence, and variance-growth primitives.

C.1 Variance Consistency Under Stabilization

When the conditional variance stabilizes, we obtain a conventional CLT.

Assumption C.1 (Variance stabilization)**.** There exists $V \in (0, \infty)$ such that $V_{\mathcal{T}}^2/n_{\text{eff}} \xrightarrow{p} V$.

Corollary C.2 (Non-studentized CLT under variance stabilization)**.** *Assume Assumptions 3.6, 3.2, 3.7, 3.8, 4.5, 4.8, 3.9, 3.10, 5.11, and C.1. Then* $\sqrt{n_{\text{eff}}}(\widehat{\theta} - \theta_0) \Rightarrow \mathcal{N}(0, V)$ *and* $\widehat{V} \to_p V$.

Proof. By Theorem 5.14,

$$\frac{\sqrt{n_{\text{eff}}}(\widehat{\theta} - \theta_0)}{\sqrt{\widehat{V}}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Moreover the proof of Theorem 5.14 shows both $Q_{\mathcal{T}}/V_{\mathcal{T}}^2 \xrightarrow{p} 1$ and $(n_{\text{eff}} - 1)\widehat{V}/Q_{\mathcal{T}} \xrightarrow{p} 1$, hence $\widehat{V} = (V_{\mathcal{T}}^2/n_{\text{eff}}) \cdot (1 + o_{\mathbb{P}}(1))$. Under Assumption C.1, $V_{\mathcal{T}}^2/n_{\text{eff}} \xrightarrow{p} V$, so $\widehat{V} \xrightarrow{p} V$ and Slutsky yields $\sqrt{n_{\text{eff}}}(\widehat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V)$. □

C.2 Oracle Equivalence

Under additional nuisance consistency, the feasible estimator is asymptotically equivalent to the oracle.

Assumption C.3 (L^2 nuisance consistency)**.** Let $m_a^\star(x) := \mathbb{E}[Y(a) \mid X = x]$. Then, for each $a \in \{0, 1\}$, we have (where the expectation is over the full data stream and any learner randomness used to construct $\widehat{m}$)

$$\frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{T}} \mathbb{E}\Big[\big(\widehat{m}_{t-1,a}(X_t) - m_a^\star(X_t)\big)^2\Big] \to 0.$$

Theorem C.4 (Oracle equivalence)**.** *Assume Assumptions 3.6, 3.2, 3.7, 3.8, 4.5, and C.3. Assume also the scored set* $\mathcal{T}$ *is deterministic (Assumption 4.8). Let* $m^\star := (m_0^\star, m_1^\star)$ *and define the oracle estimator*

$$\widehat{\theta}^\star := \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{T}} \phi_t(m^\star).$$

Then

$$\sqrt{n_{\text{eff}}}(\widehat{\theta} - \widehat{\theta}^\star) \xrightarrow{p} 0.$$

Proof. Let $e_{t-1,a}(x) := \widehat{m}_{t-1,a}(x) - m_a^\star(x)$. A direct algebraic calculation gives, for each scored index $t \in \mathcal{T}$,

$$\phi_t(\widehat{m}_{t-1}) - \phi_t(m^\star) = -(A_t - \pi_t)\left(\frac{e_{t-1,1}(X_t)}{\pi_t} + \frac{e_{t-1,0}(X_t)}{1 - \pi_t}\right) =: \Delta_t.$$

By predictability, $(e_{t-1,0}, e_{t-1,1})$ is $\mathcal{F}_{t-1}$-measurable, and π_t is $\mathcal{G}_t$-measurable. Using iterated expectations,

$$\mathbb{E}[\Delta_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\mathbb{E}[\Delta_t \mid \mathcal{G}_t] \mid \mathcal{F}_{t-1}] = -\mathbb{E}\left[\left(\frac{e_{t-1,1}(X_t)}{\pi_t} + \frac{e_{t-1,0}(X_t)}{1 - \pi_t}\right)\mathbb{E}[A_t - \pi_t \mid \mathcal{G}_t] \,\middle|\, \mathcal{F}_{t-1}\right] = 0,$$

so $\{\Delta_t, \mathcal{F}_t\}$ is a martingale difference sequence over $t \in \mathcal{T}$. Now

$$\sqrt{n_{\text{eff}}}\,(\widehat{\theta} - \widehat{\theta}^\star) = \frac{1}{\sqrt{n_{\text{eff}}}} \sum_{t \in \mathcal{T}} \Delta_t.$$

By orthogonality of martingale differences,

$$\mathbb{E}\left[\left(\sqrt{n_{\text{eff}}}\,(\widehat{\theta}-\widehat{\theta}^{\star})\right)^2\right] = \frac{1}{n_{\text{eff}}}\sum_{t\in\mathcal{T}}\mathbb{E}[\Delta_t^2].$$

Conditional on $\mathcal{G}_t$, $\mathbb{E}[(A_t-\pi_t)^2 \mid \mathcal{G}_t] = \pi_t(1-\pi_t)$, hence

$$\mathbb{E}[\Delta_t^2 \mid \mathcal{G}_t] = \pi_t(1-\pi_t)\left(\frac{e_{t-1,1}(X_t)}{\pi_t} + \frac{e_{t-1,0}(X_t)}{1-\pi_t}\right)^2 \le C\big(e_{t-1,1}(X_t)^2 + e_{t-1,0}(X_t)^2\big)$$

for a constant $C<\infty$ depending only on the overlap constant ε. Taking expectations and averaging over $t\in\mathcal{T}$ yields

$$\mathbb{E}\left[\left(\sqrt{n_{\text{eff}}}\,(\widehat{\theta}-\widehat{\theta}^{\star})\right)^2\right] \le \frac{C}{n_{\text{eff}}}\sum_{t\in\mathcal{T}}\mathbb{E}\big[e_{t-1,1}(X_t)^2 + e_{t-1,0}(X_t)^2\big] \to 0$$

by Assumption C.3. Therefore $\sqrt{n_{\text{eff}}}(\widehat{\theta}-\widehat{\theta}^{\star}) \xrightarrow{p} 0$. □

Remark C.5 (Relation to i.i.d. DML rate conditions). In i.i.d. DML analyses of the ATE with *unknown* propensities, asymptotic linearity typically requires a product-rate condition between the propensity-score error and the outcome-regression error (often enforced via $o_P(n^{-1/4})$-type rates for each component); see, e.g., (Chernozhukov et al., 2018, 2022). In our setting, propensities are assumed known and logged, so the remaining nuisance component enters only through predictable regression adjustments. As a result, nuisance convergence is not needed for validity of the studentized CLT, and L^2 consistency is sufficient for oracle equivalence (Theorem C.4).

C.3 A Sufficient Condition for Variance Growth

Assumption 5.11 is mild but abstract. The following provides a primitive sufficient condition.

Assumption C.6 (Nontrivial noise)**.** $\mathbb{E}[\sigma_1^2(X)+\sigma_0^2(X)] \ge \underline{\sigma}^2 > 0$.

Lemma C.7 (A sufficient condition for variance growth)**.** *Assume Assumptions 3.6, 3.2, 3.7, 3.8, and C.6. Let $\xi_t := \phi_t(\widehat{m}_{t-1}) - \theta_0$ with $\widehat{m}_{t-1}$ satisfying Assumption 4.5, and let $V_{\mathcal{T}}^2 := \sum_{t\in\mathcal{T}}\mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}]$ as in (5.3). Then there exists $v_- > 0$ such that for every scored set $\mathcal{T}$ with $n_{\text{eff}} := |\mathcal{T}| \to \infty$,*

$$\mathbb{P}\big(V_{\mathcal{T}}^2 \ge v_-\, n_{\text{eff}}\big) \to 1,$$

so Assumption 5.11 holds.

Proof. Fix $t\in\mathcal{T}$. By Proposition 5.8 and nonnegativity of the augmentation term,

$$\mathbb{E}[\xi_t^2 \mid \mathcal{G}_t] \ge \frac{\sigma_1^2(X_t)}{\pi_t} + \frac{\sigma_0^2(X_t)}{1-\pi_t}.$$

By overlap (Assumption 3.8), $\pi_t \le 1-\varepsilon$ and $1-\pi_t \le 1-\varepsilon$ (so $1/\pi_t \ge 1/(1-\varepsilon)$ and $1/(1-\pi_t) \ge 1/(1-\varepsilon)$), hence

$$\frac{\sigma_1^2(X_t)}{\pi_t} + \frac{\sigma_0^2(X_t)}{1-\pi_t} \ge \frac{\sigma_1^2(X_t)+\sigma_0^2(X_t)}{1-\varepsilon}.$$

Assumption	Informal description	Used in
3.5	SUTVA / no interference	Identification of θ_0 (Section 3); standing throughout.
3.6	i.i.d. arrivals / no selection (or mean-stationarity alternative)	Lem. 5.3; Prop. 5.8; Lem. C.7 (Appendix C.3); Thm. 5.14 via Lem. 5.3.
3.2	logged *executed* propensities	Lem. 5.3; Prop. 5.8; Lem. C.7.
3.7	sequential randomization / no unmeasured confounding	Lem. 5.3; Prop. 5.8; Lem. C.7.
4.8	prespecified (deterministic) scored set $\mathcal{T}$	Thm. 5.14 (via predictable indicator $\mathbb{1}\{t \in \mathcal{T}\}$); Appendix B proof of Thm. 5.14.
4.5	nuisances used for unit t depend only on past	Lem. 5.3; Thm. 5.14.
3.8	executed overlap on scored units	Lem. 5.6; Lem. C.7; Thm. 5.14 via Lem. 5.6.
3.9	bounded moments of potential outcomes	Lem. 5.6; Thm. 5.14.
3.10	stability of nuisance fits (bounded moments)	Lem. 5.6; Thm. 5.14.
5.11	variance growth ($V_{\mathcal{T}}^2 \to \infty$)	Thm. 5.14 (verifying conditions of Thm. A.1).
C.6	nondegenerate outcome noise	Cor. 5.12; Lem. C.7.
C.1	deterministic variance limit (stabilization)	Cor. C.2.

Table 10: Assumption-to-result map

Taking $\mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$ and using i.i.d. arrivals (Assumption 3.6) yields

$$\mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}] = \mathbb{E}\big[\mathbb{E}[\xi_t^2 \mid \mathcal{G}_t] \mid \mathcal{F}_{t-1}\big] \geq \frac{1}{1-\varepsilon}\mathbb{E}[\sigma_1^2(X) + \sigma_0^2(X)] \geq \frac{\underline{\sigma}^2}{1-\varepsilon} =: v_-,$$

almost surely. Summing over $t \in \mathcal{T}$ gives $V_{\mathcal{T}}^2 = \sum_{t\in\mathcal{T}} \mathbb{E}[\xi_t^2 \mid \mathcal{F}_{t-1}] \geq v_- n_{\text{eff}}$ almost surely, hence the probability statement holds. □

D Additional simulations

To keep this methods note self-contained and reproducible from the paper text, we omit additional high-dimensional contextual-bandit simulations beyond Designs A–C.

E Assumption-to-Result Map

F Data contract and operational logging requirements

This appendix describes a minimal "data contract" for implementing Theorem 5.14 in an experimentation platform. The contract separates (i) *auditable* pipeline conditions (executed-propensity logging and predictable nuisance construction) from (ii) substantive causal/model assumptions (e.g., SUTVA/no interference, sequential randomization, and no selection) that are not fully verifiable from logs alone.

Persisted artifact	Audit / validation check
Executed propensity π_t and assignment record for each unit t (including any clipping rule and RNG call metadata)	**Executed propensity integrity.** Verify that the logged π_t is the probability passed to the randomization device. As a coarse calibration diagnostic, compare $\overline{A}_B$ to $\overline{\pi}_B$ within propensity bins B; this can catch severe implementation errors, but it does not certify correct logging/randomization or validate the causal assumptions (Remark 3.3).
Experiment-level overlap rule (e.g., clipping threshold ε) and the realized min/max of π_t on scored units	**Executed overlap.** Check that $\pi_t \in [\varepsilon, 1-\varepsilon]$ for all $t \in \mathcal{T}$. If overlap is enforced by clipping, confirm clipping occurred at assignment time.
Scored-set definition and indicator $T_t := \mathbb{1}\{t \in \mathcal{T}\}$ (or a documented predictable rule)	**Scored-set integrity.** Confirm $\mathcal{T}$ is prespecified (Assumption 4.8) or, if a predictable rule is used (e.g., due to outcome delay), document the rule and verify it does not depend on future outcomes.
Nuisance-fit artifacts for each scoring block (training indices, timestamps, learner configuration, hyperparameters, and randomness/seeds)	**Predictability.** Verify that the nuisance fit used to score unit t was trained only on indices $< t$, and that the nuisance is held fixed while scoring all units in its associated block (Definition 4.3 and Assumption 4.5).
Prespecified reporting horizon n (and, if applicable, an analysis-query log)	**Fixed horizon / no optional stopping.** Confirm that confidence intervals are produced only at the prespecified horizon(s). If the experiment is continuously monitored or stopped adaptively, the fixed-horizon interval is not guaranteed; use time-uniform methods instead (Remark 5.16).

Table 11: Minimal logging contract and audit checks for fixed-horizon inference.

Design-time overlap vs analysis-time truncation. Overlap should be enforced at the *assignment* stage if needed (e.g., by clipping π_t before randomization). Post-hoc truncation of inverse-propensity weights in the analysis generally changes the estimand and can introduce bias; if units are dropped or trimmed based on realized propensities, the target estimand must be redefined explicitly.

Extension to multi-arm treatments. For $K > 2$ arms with assignment probabilities $\pi_{t,a}$, the same contract applies: log the executed $\pi_{t,a}$ for all arms, enforce $\pi_{t,a} \geq \varepsilon$ on scored units, and construct nuisance fits predictably. The AIPW score and studentizer are obtained by replacing binary inverse-propensity terms by their K-arm analogues.

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